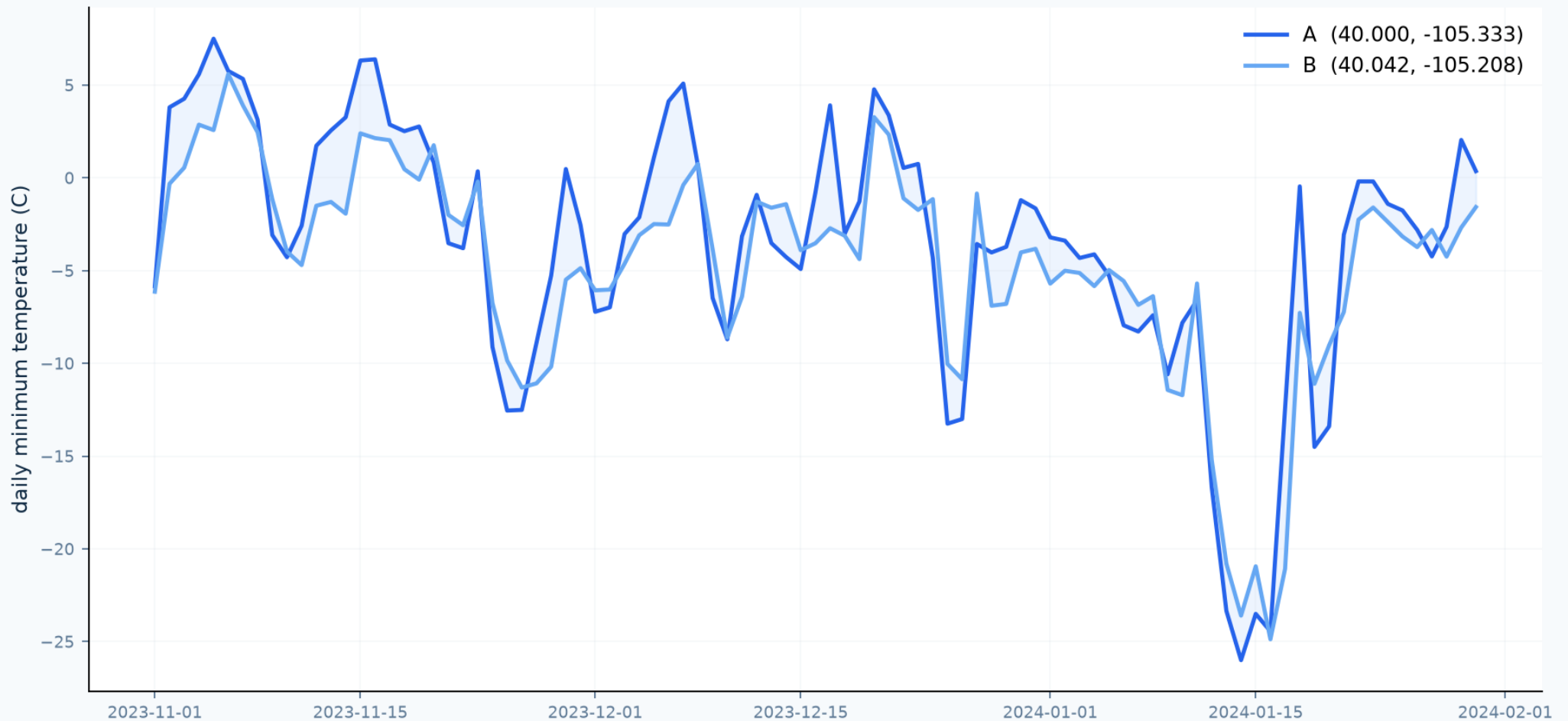


When do two places experience climate together?

One real Front Range pair, followed through the same 90-day analysis window

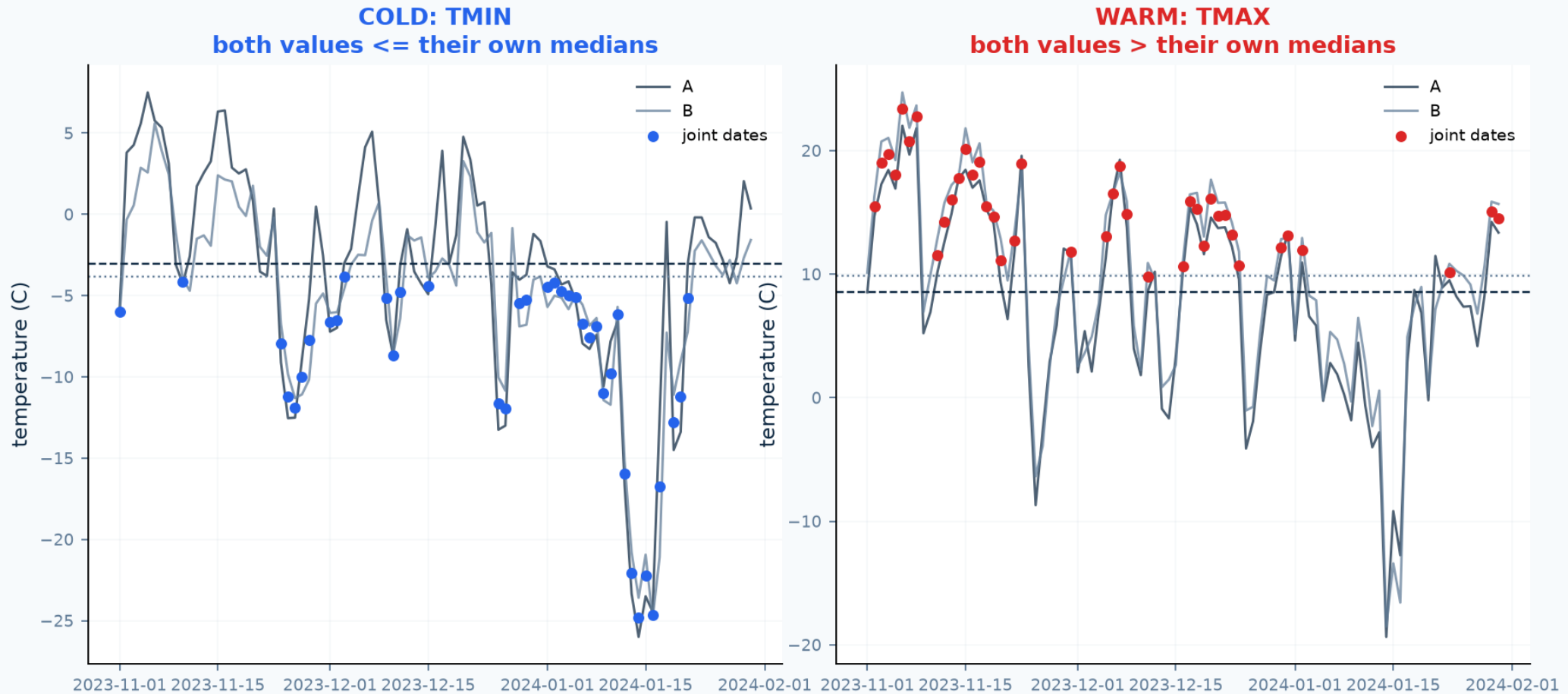


Not: Are all dates correlated?

Instead: How do they move together in cold and warm tails?

What counts as cold or warm?

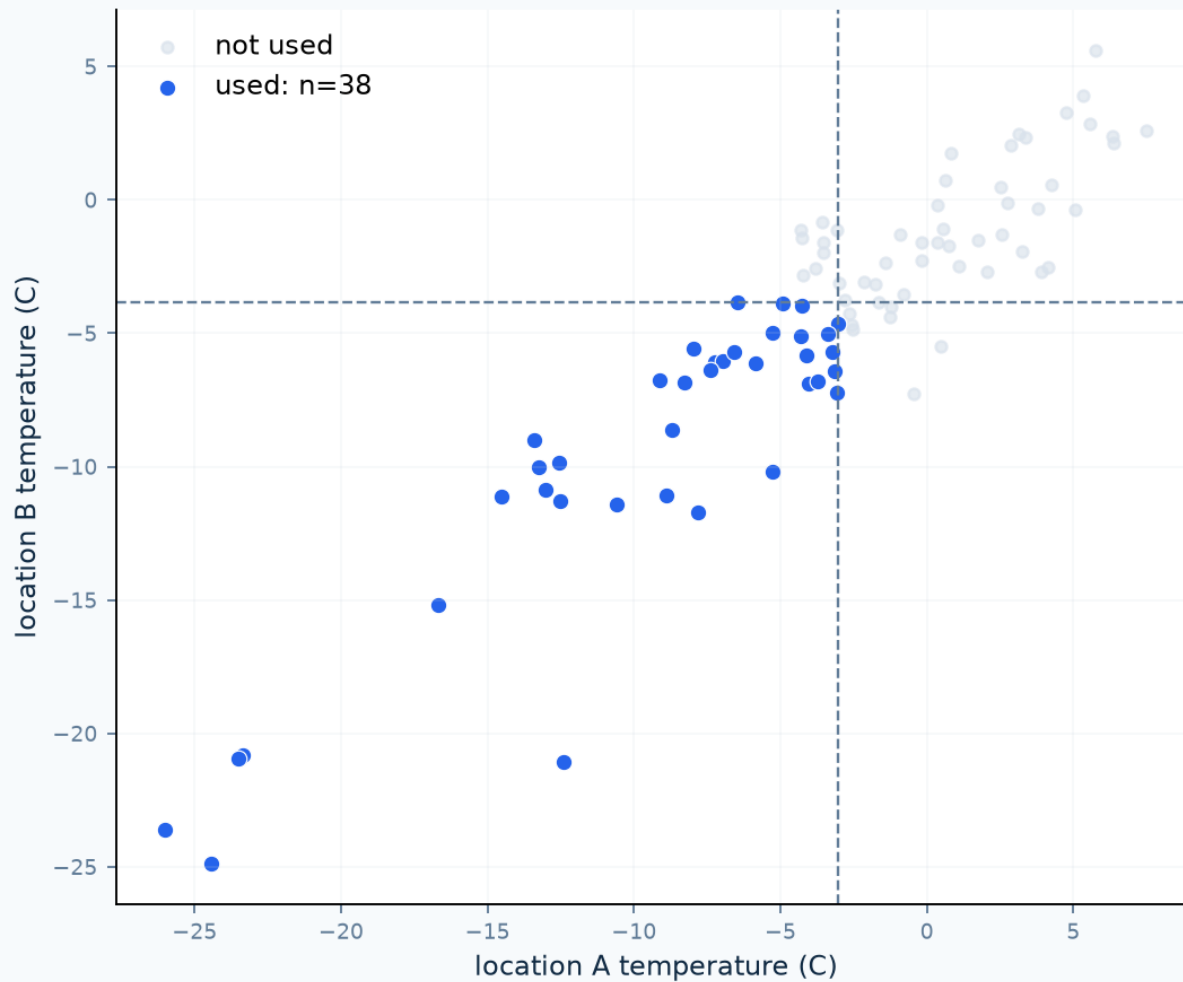
Current CubeDynamics semantics: per-location medians on pair-valid observations



The threshold is relative to each location - not one shared Celsius cutoff.

Cold synchrony

Rank the values only on dates when both locations occupy the selected tail



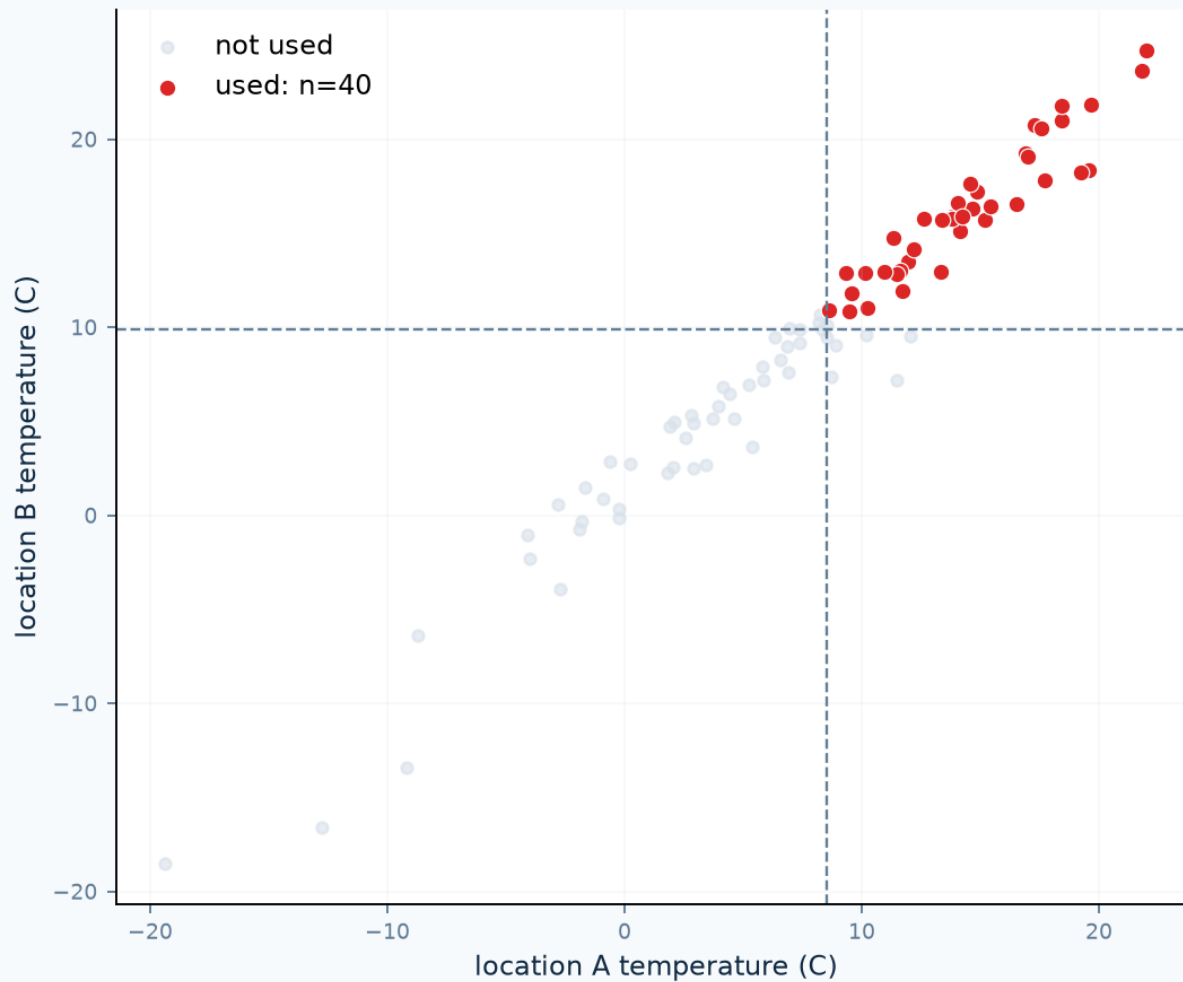
$$S_{\text{cold}}(A,B) = 0.75$$

**Spearman rank
correlation**

joint-tail dates
n = 38

Warm synchrony

Rank the values only on dates when both locations occupy the selected tail



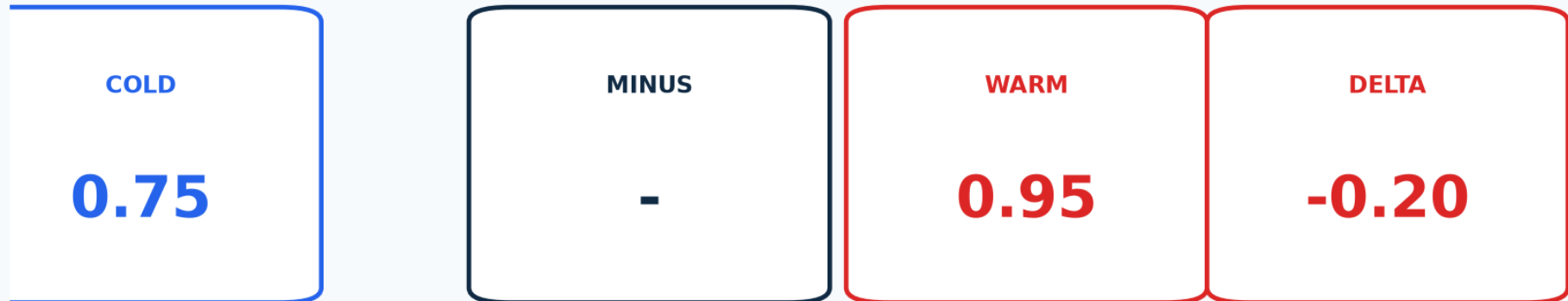
$$S_{\text{warm}}(A,B) = 0.95$$

**Spearman rank
correlation**

joint-tail dates
n = 40

Delta synchrony

The signed contrast keeps cold and warm meanings visible



$$\text{Delta } S = S_{\text{cold}} - S_{\text{warm}}$$

0

cold = warm

+

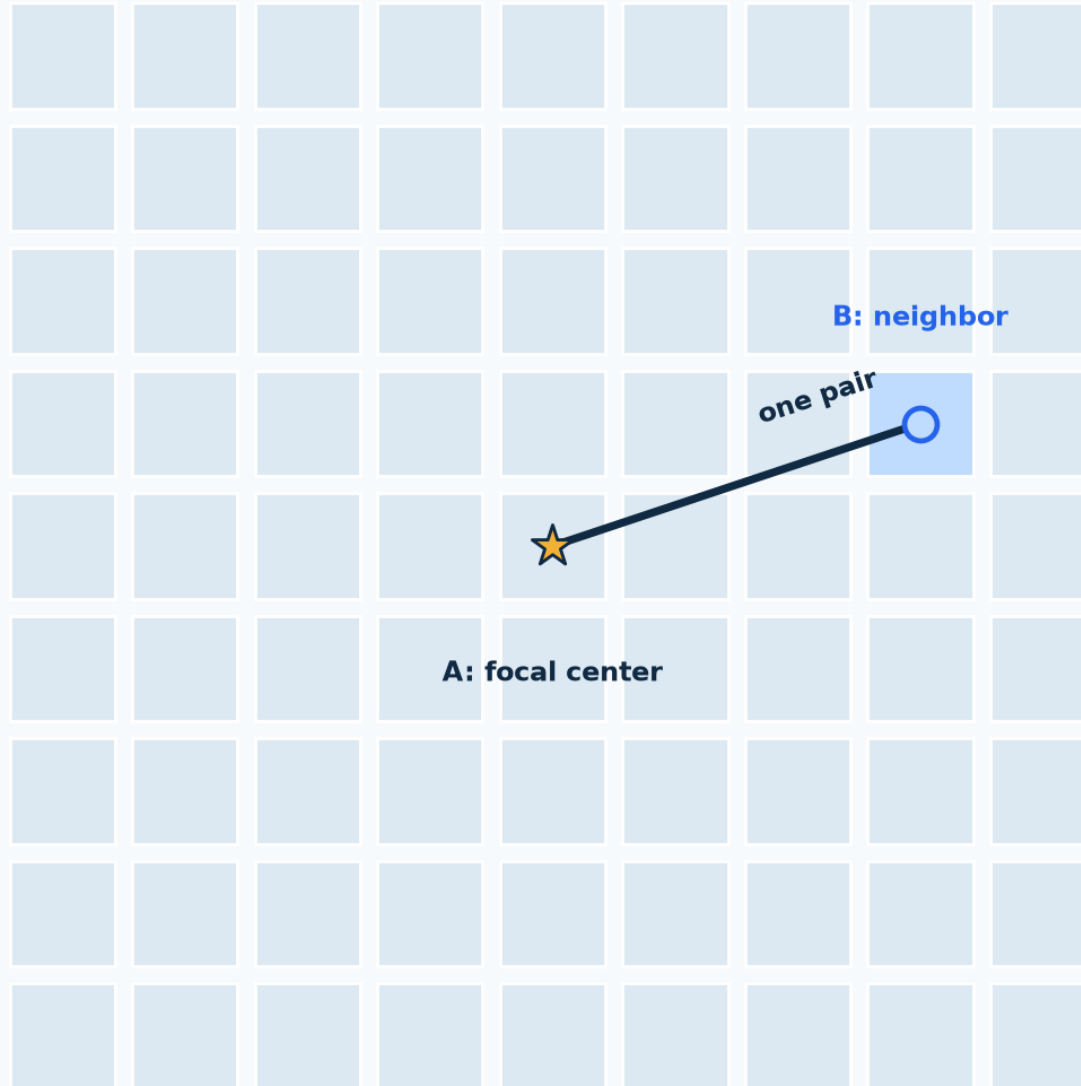
cold > warm

-

warm > cold

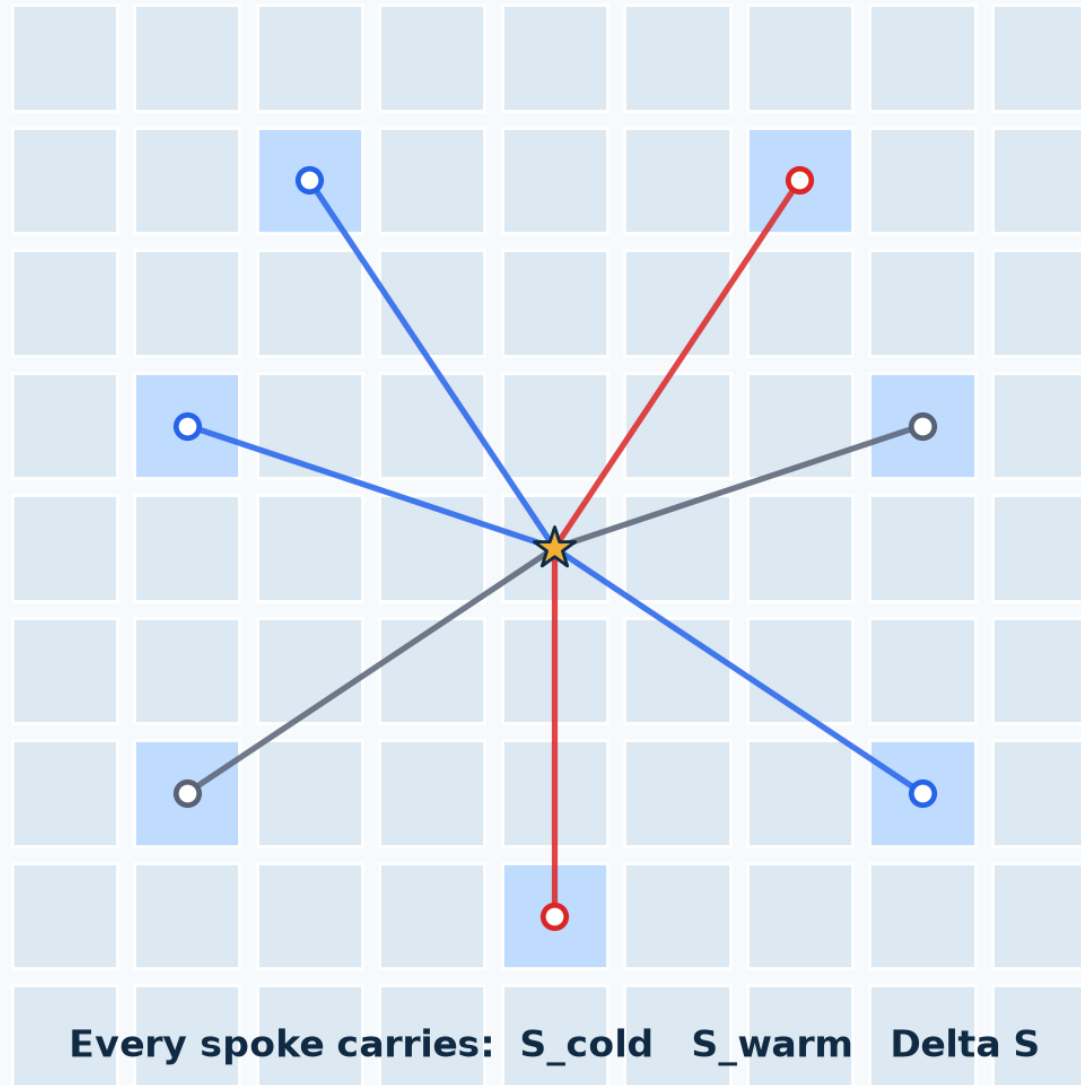
One center, one neighbor

The line is the pair relationship from the first five pages



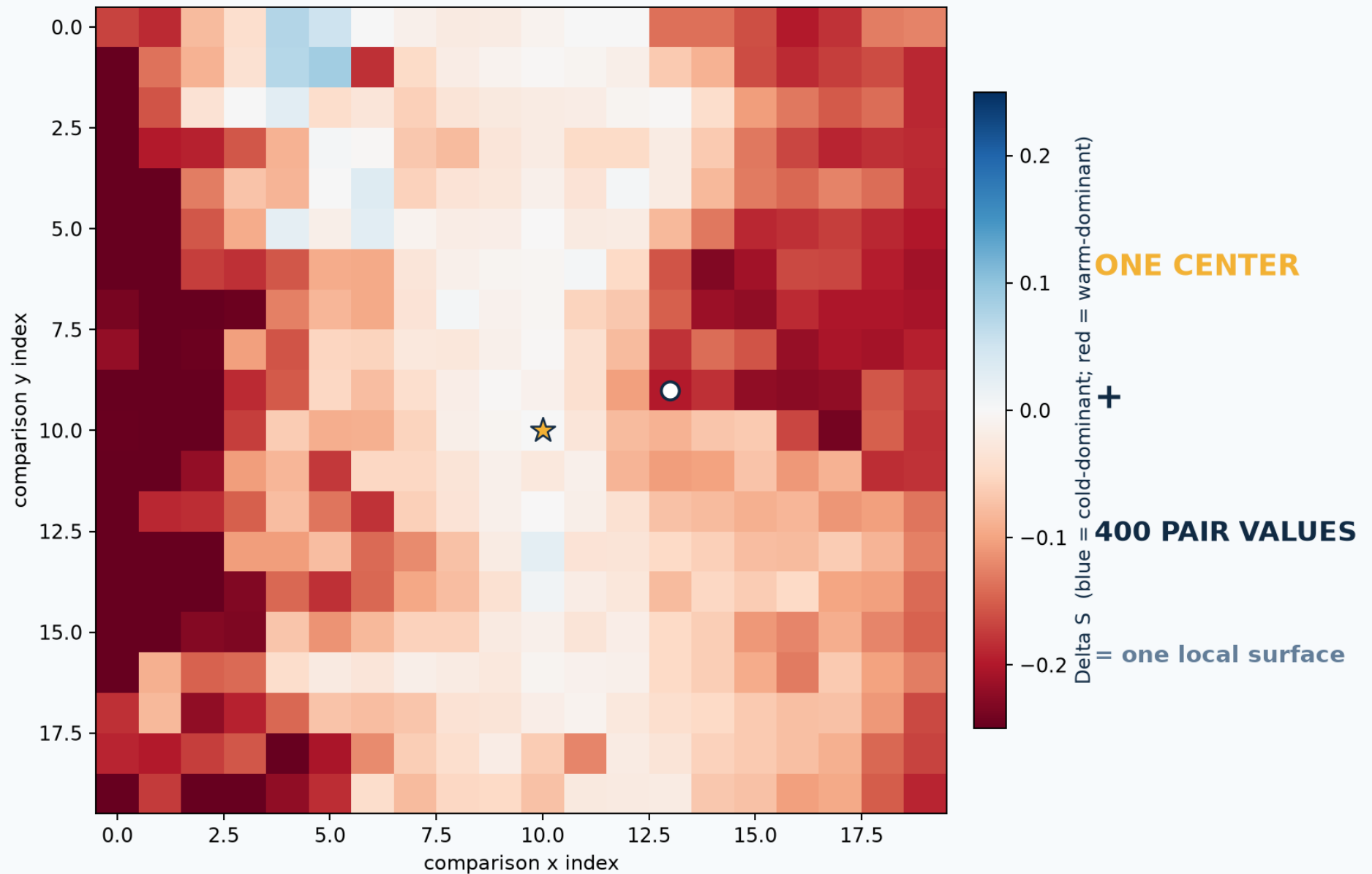
One center, many neighbors

Repeat the same pair calculation while holding the gold center fixed



The local synchrony surface

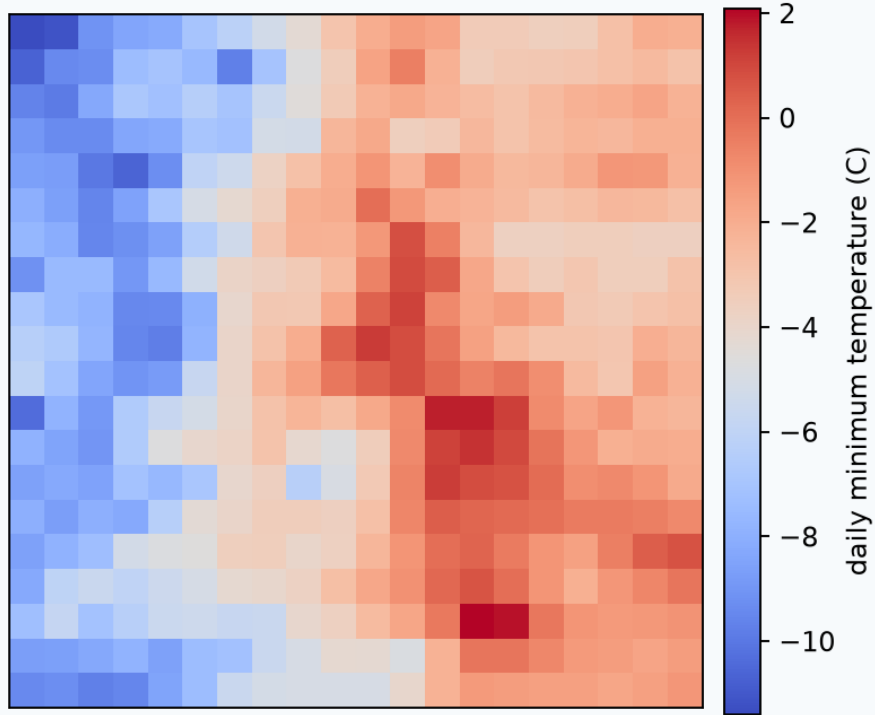
Each cell is Delta S between that cell and the SAME gold center



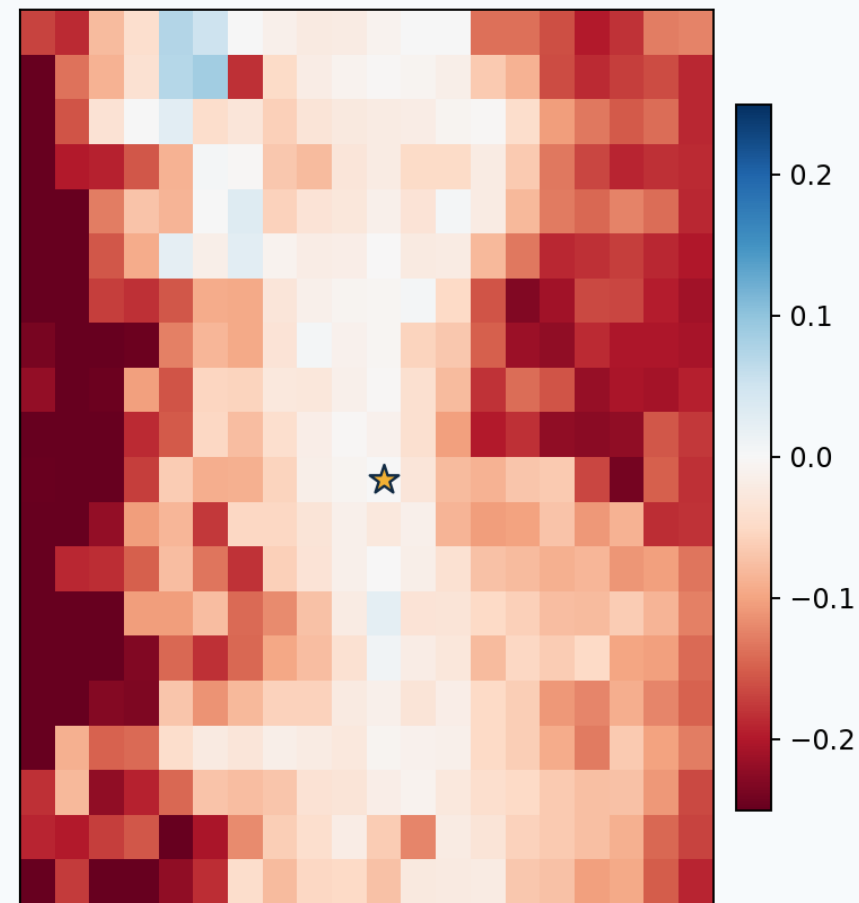
Why this is not an ordinary climate map

State at a cell versus relationship to a chosen center

Ordinary temperature map
'What is the value here?'

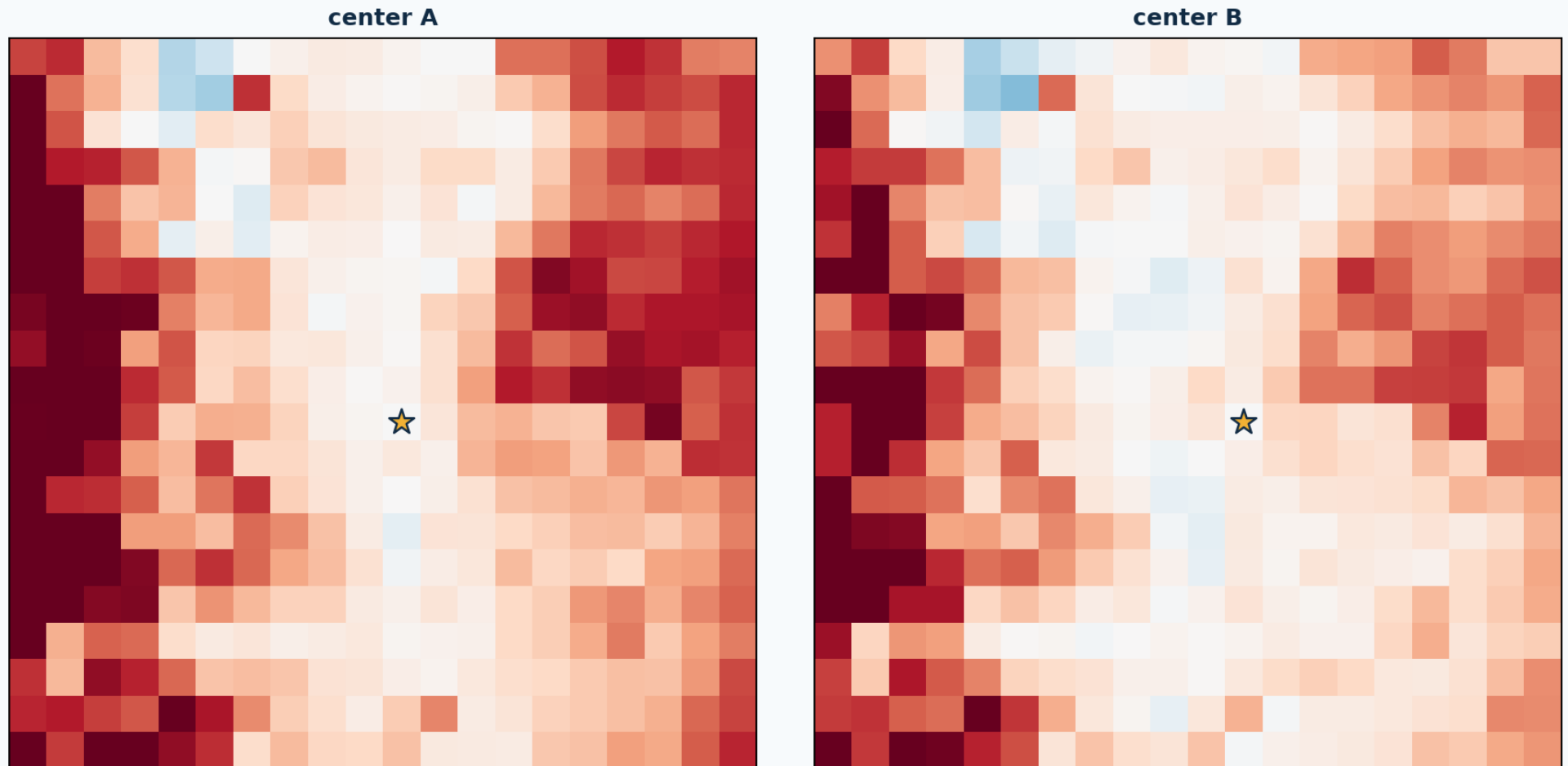


Center-reference surface
'How does here relate to the center?'



Move the center one pixel

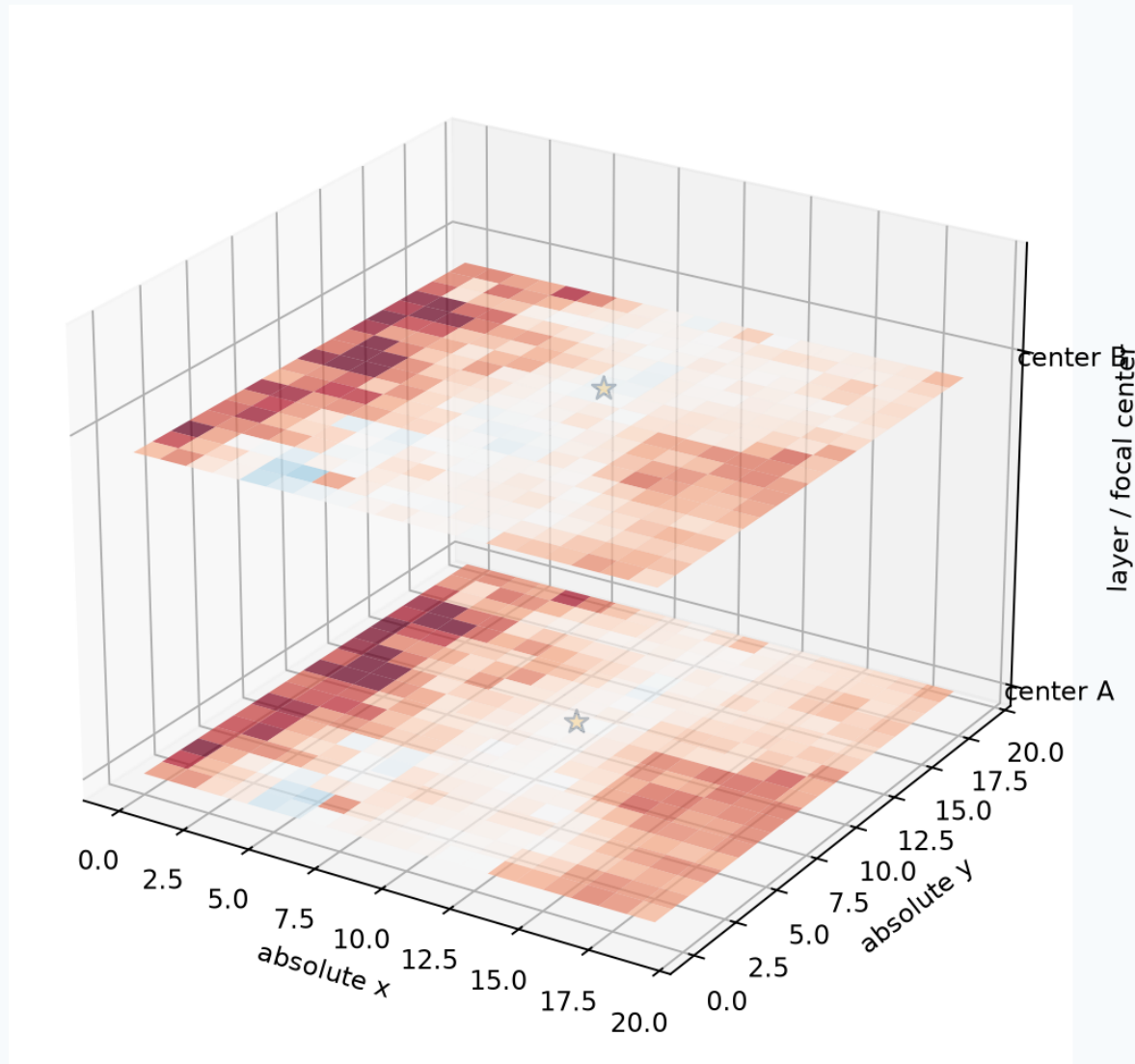
A new focal center produces a NEW local surface



Do not combine them yet: same geography, different focal relationships.

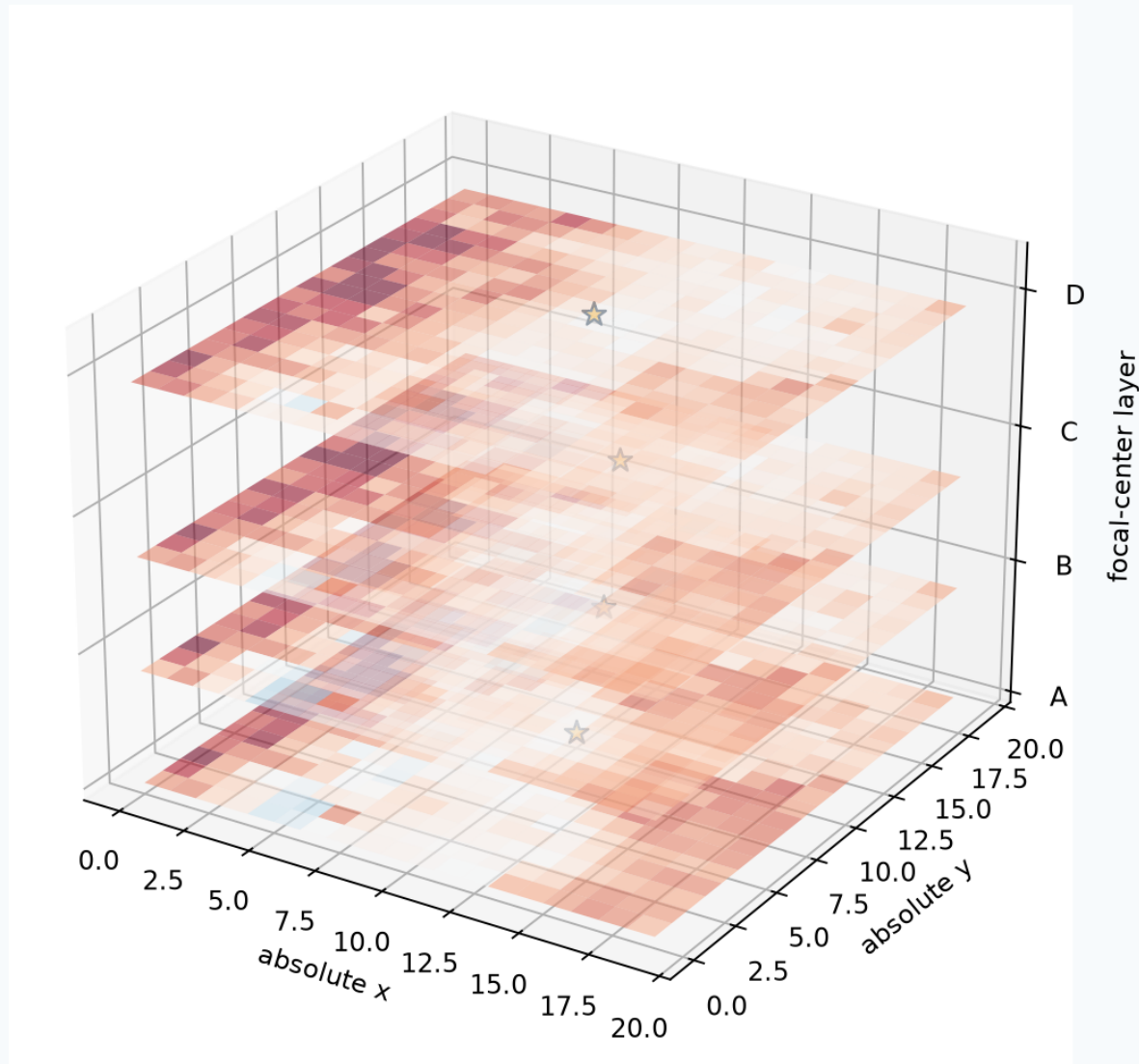
The two surfaces overlap

Layer height indexes focal center - not elevation and not time



Move again

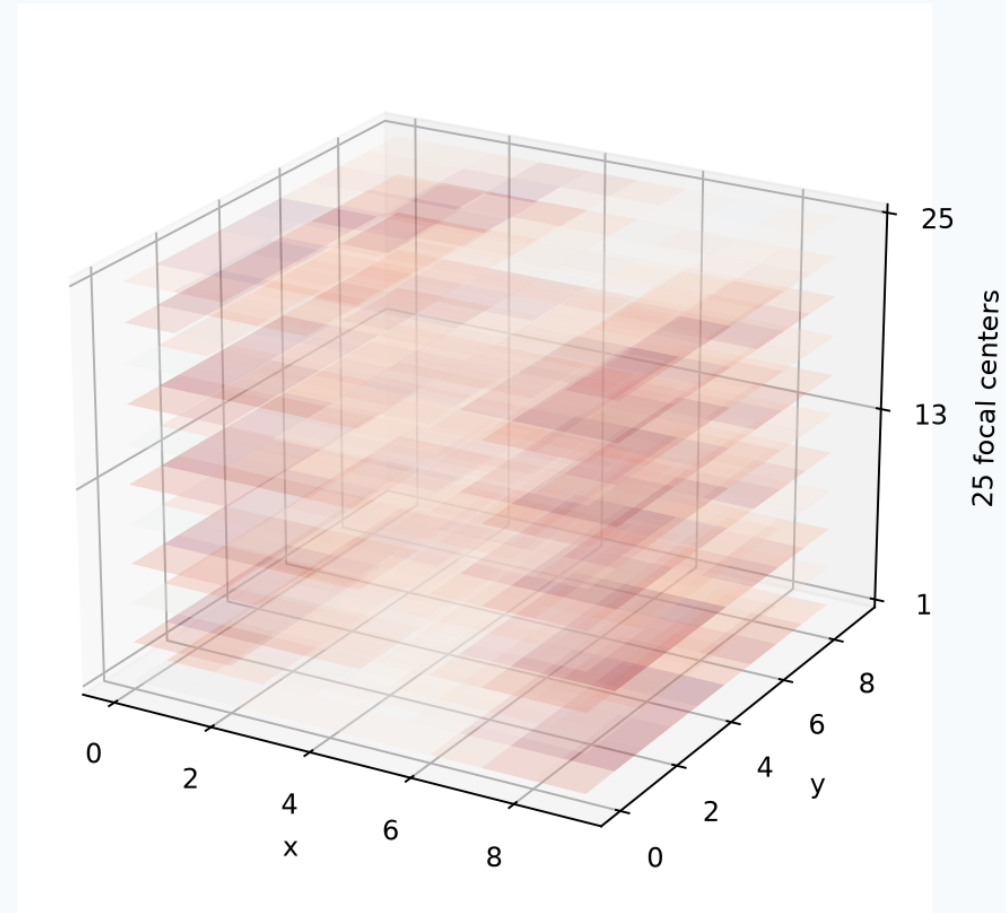
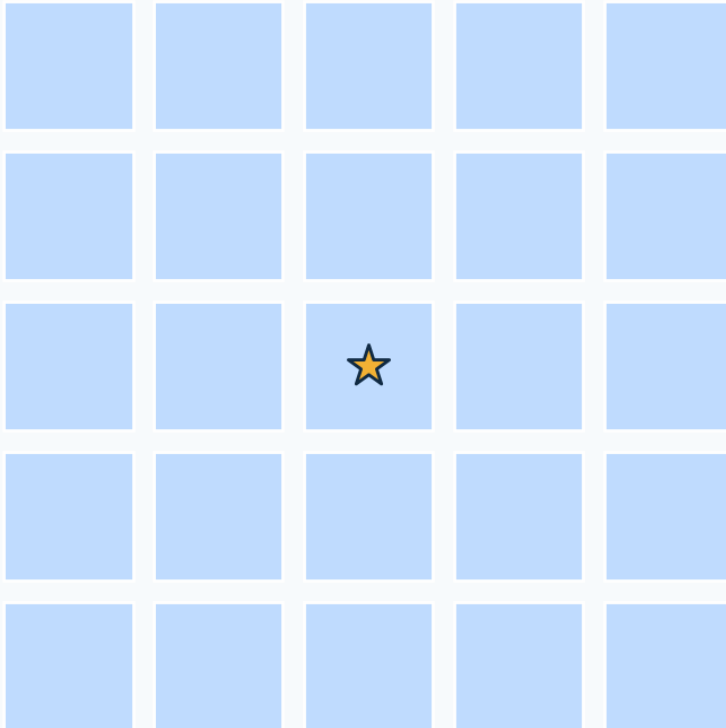
Centers A, B, C, and D each create a complete relational view



Fill a small grid of centers

25 focal pixels -> 25 overlapping relational surfaces

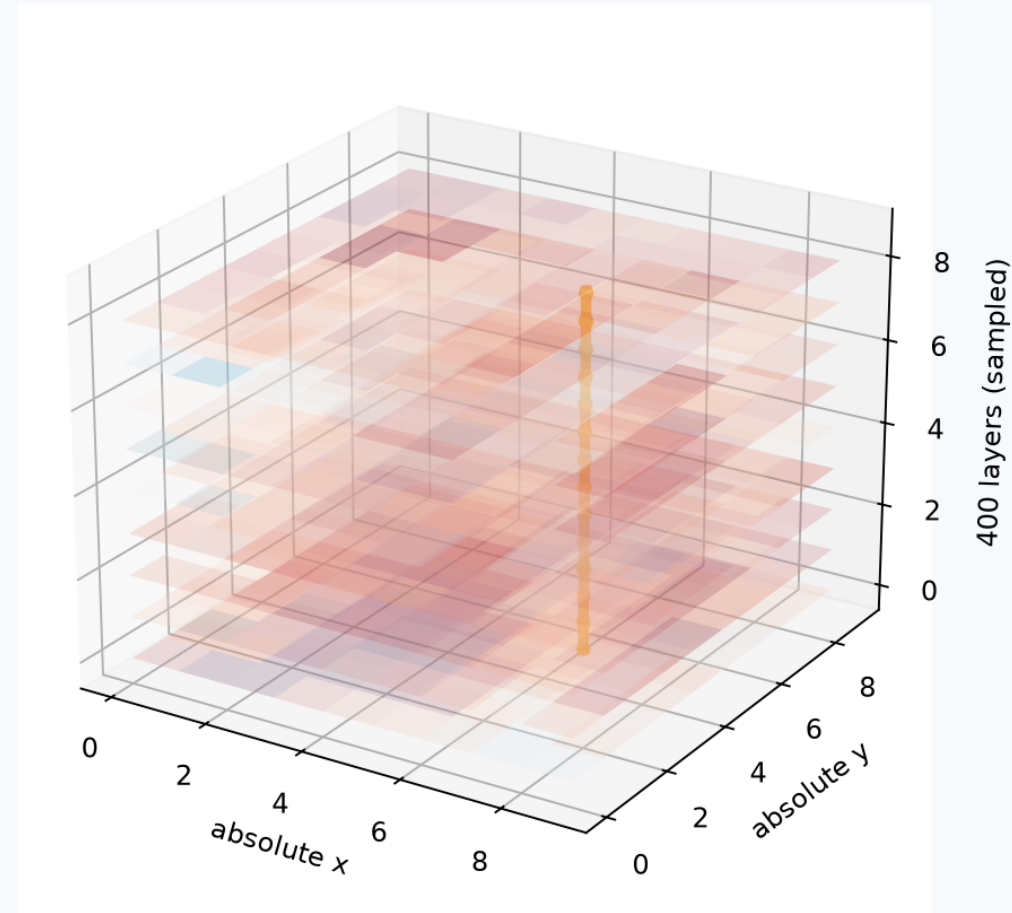
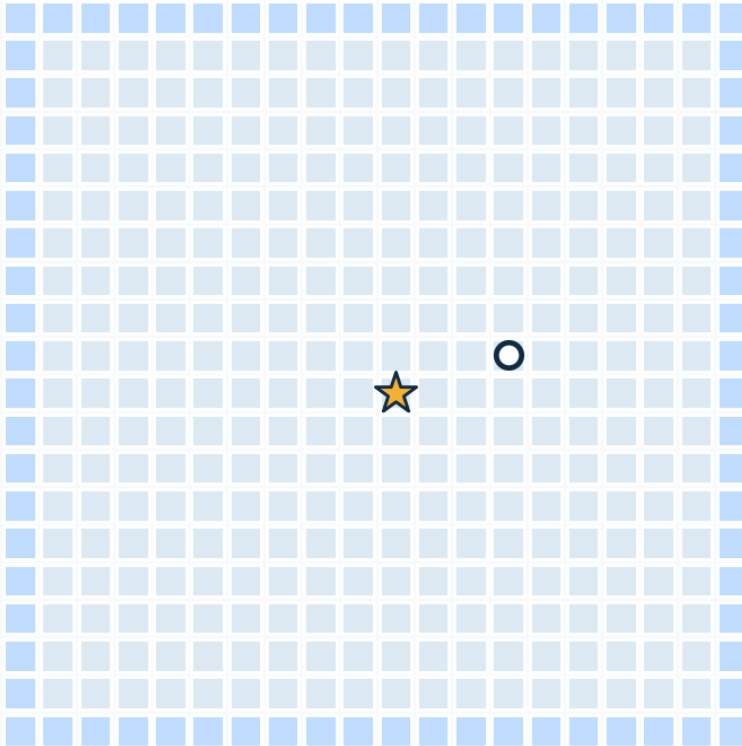
5 x 5 possible centers



The 20 x 20 thought experiment

The real block has 400 focal surfaces; one absolute cell appears in many layers

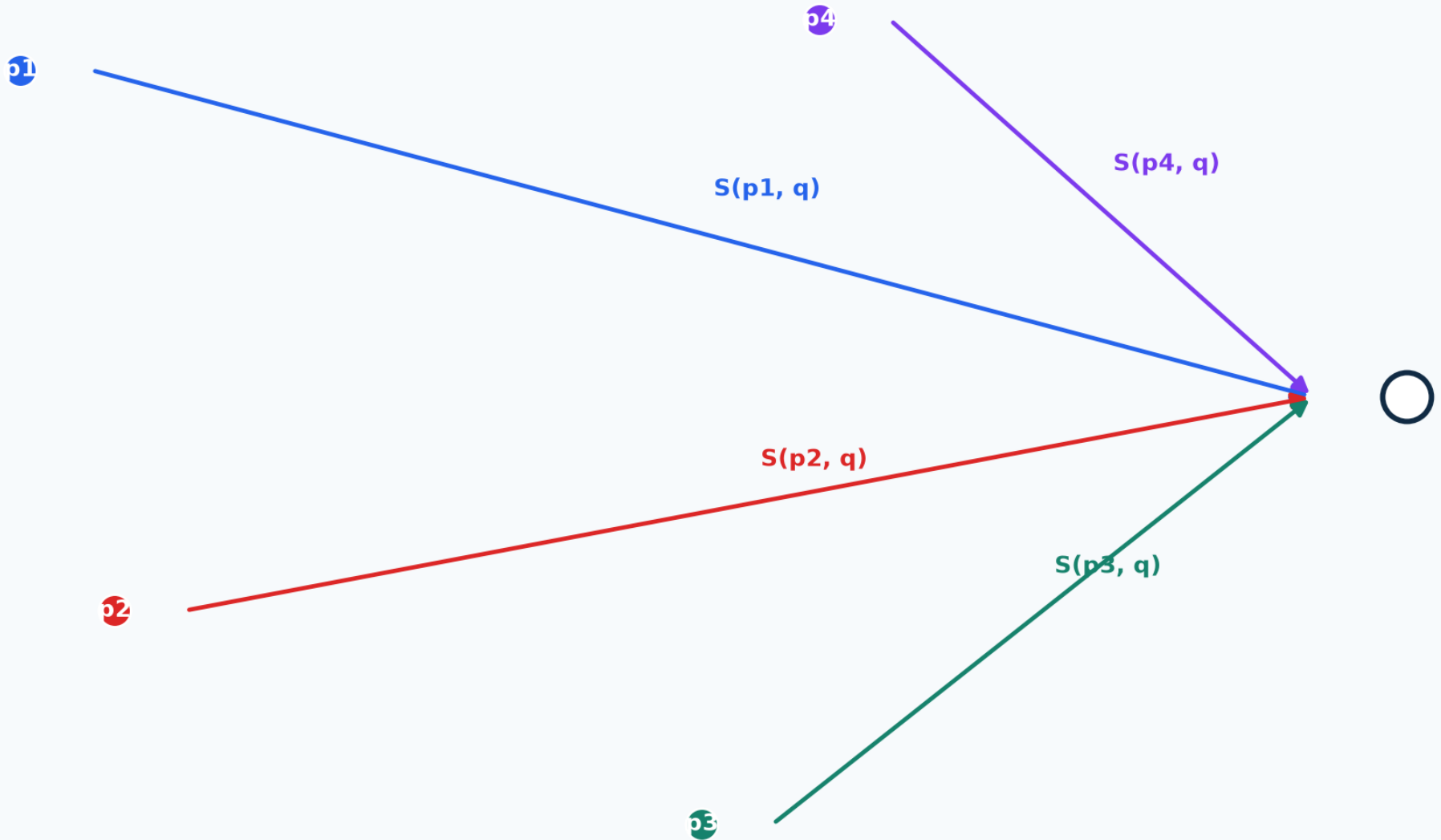
perimeter first -> fill the interior



400 surfaces x 400 cells = 160,000 directed views

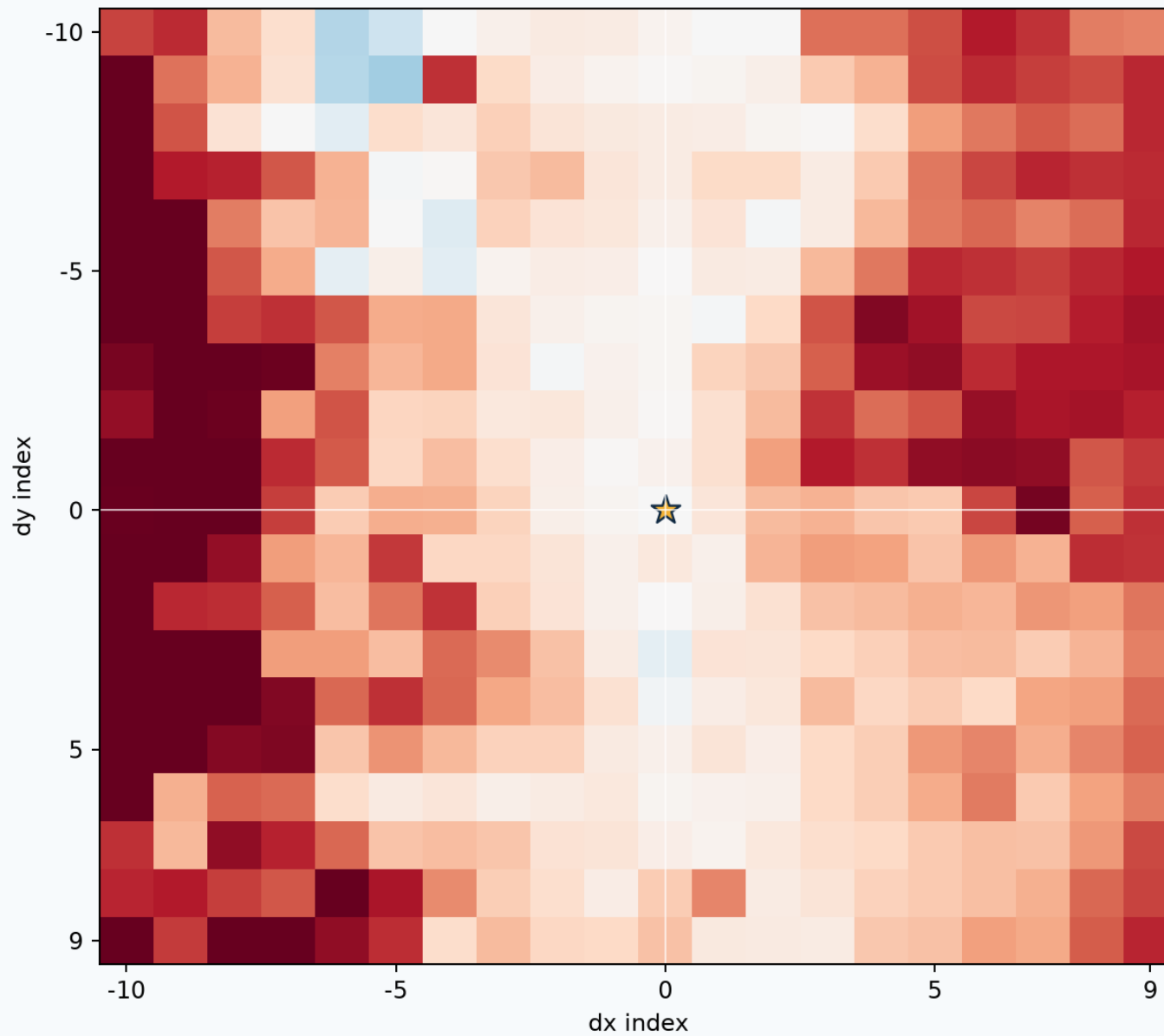
What each element in the stack means

A column through q is a collection of relationships involving q



Relative view

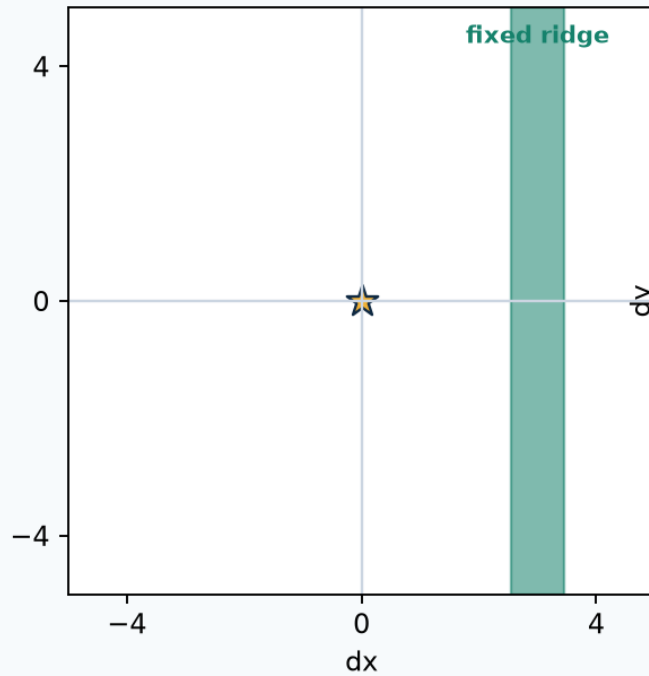
For every focal center p , describe neighbors by dx and dy ; p is always $(0,0)$



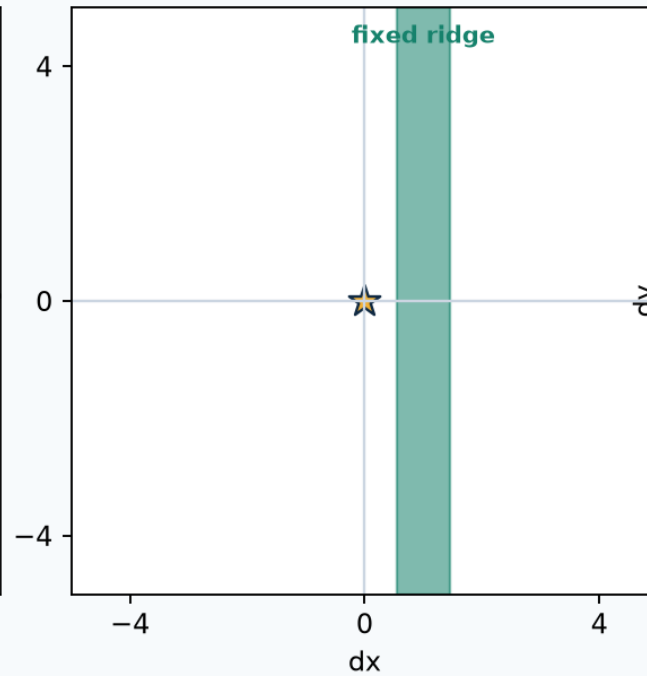
Move the center

In relative coordinates, fixed geography appears to move through the window

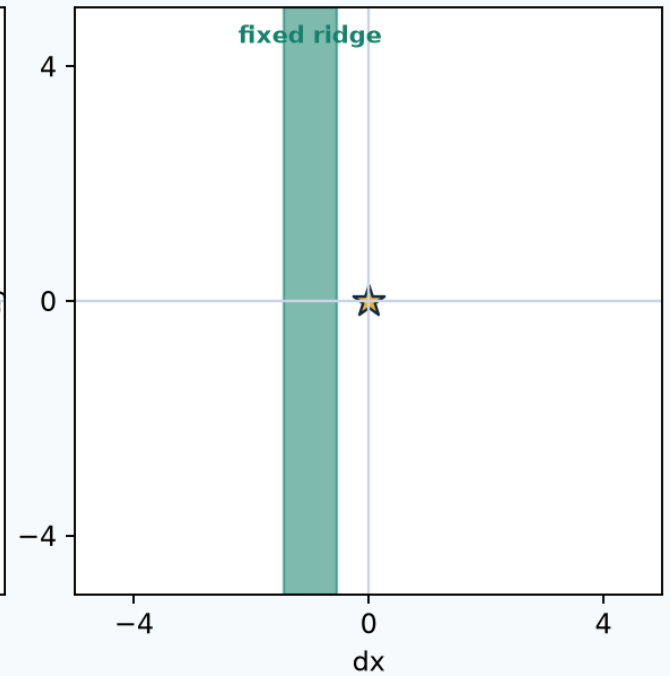
center moves east
step 1



center moves east
step 2

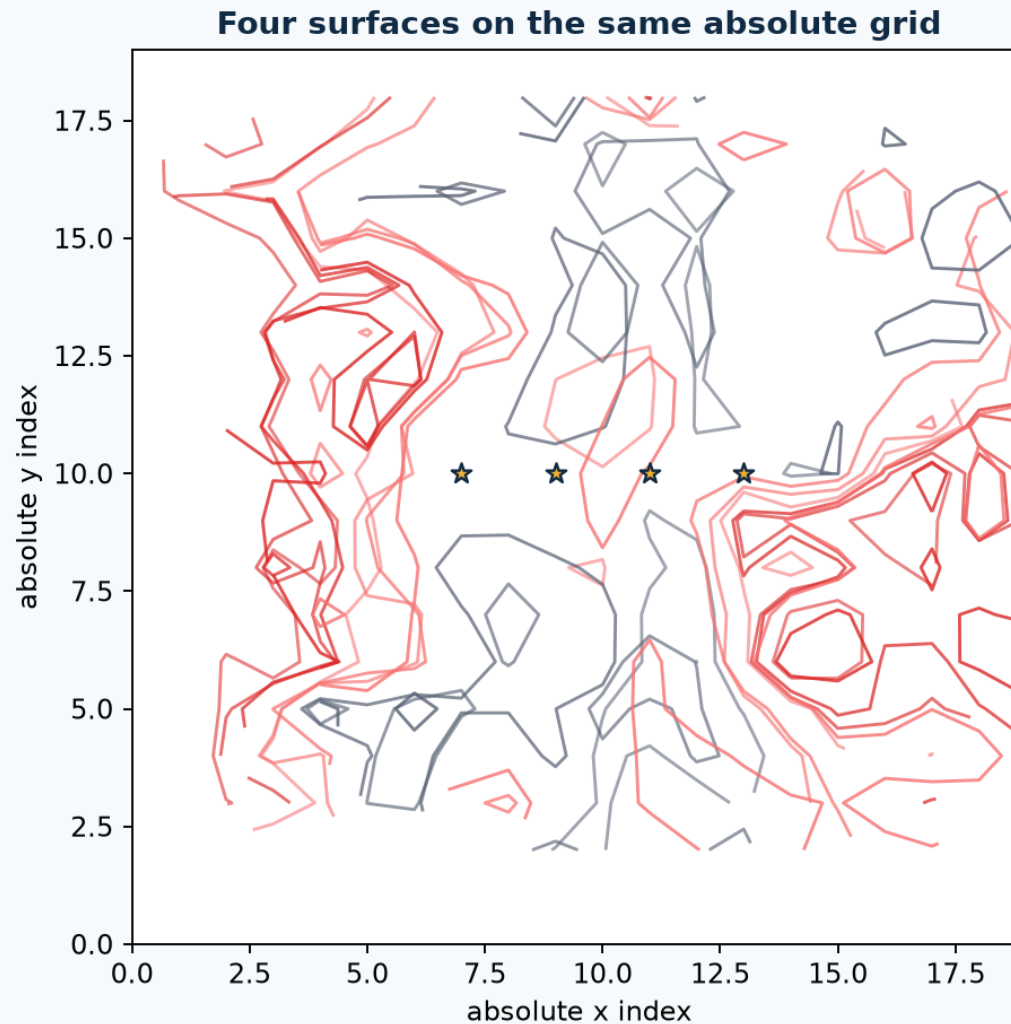


center moves east
step 3



Absolute view

Latitude/longitude keep fixed features fixed while the focal center moves

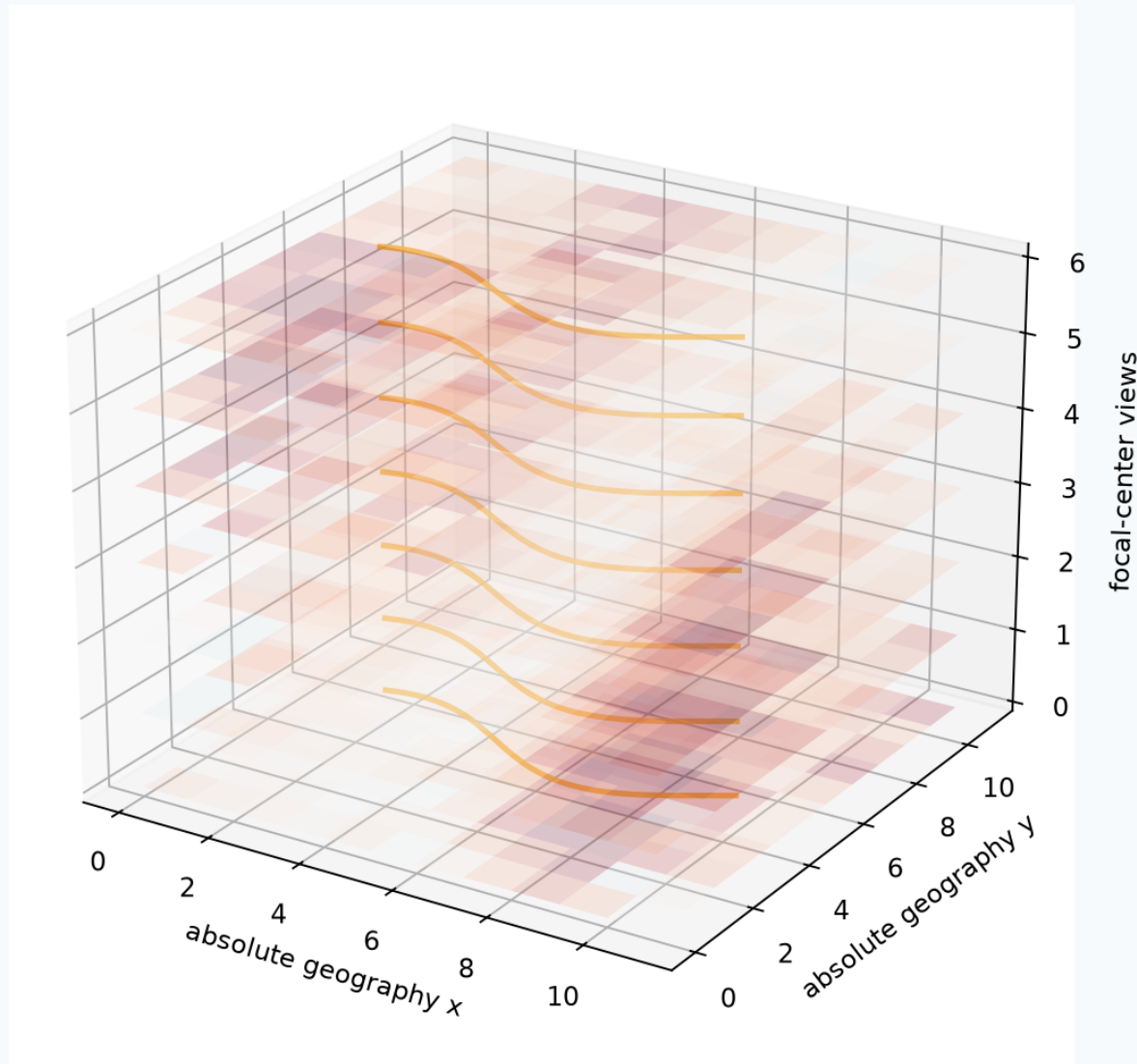


correlation
0.71 -> 0.94

gradient agreement
0.18 -> 0.93

The central insight

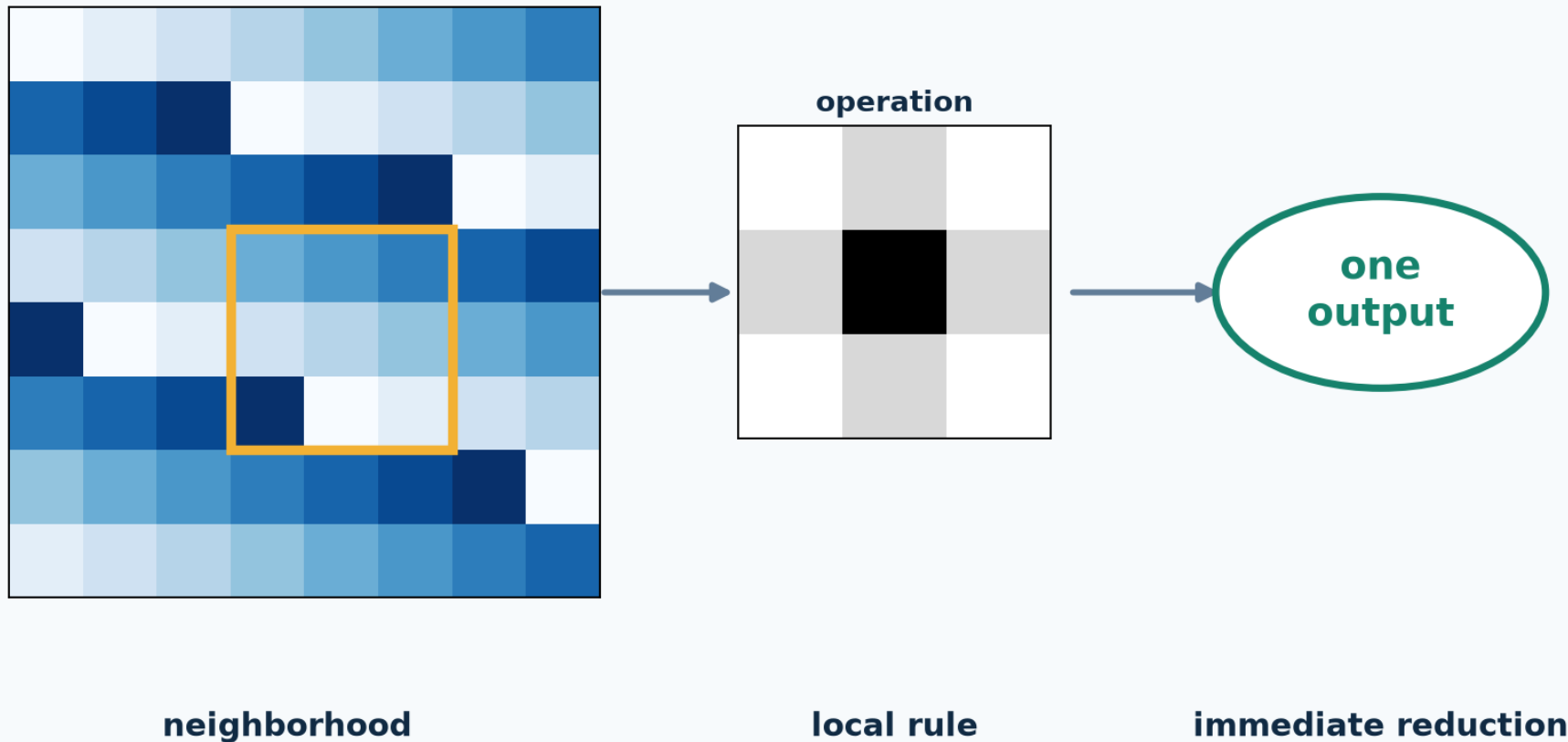
Overlap contains information only if relationship locations remain attached to geography



ALIGN FIRST. THEN ASK WHAT GEOGRAPHIC ORGANIZATION REPEATS.

Ordinary convolution

A kernel slides over an image and usually emits one reduced output per location



Our relational version

The center slides, but the scientifically defined operation returns a complete surface



difference from ordinary convolution: retain the relational neighborhood

Why this is not a CNN

The analogy describes sliding and overlapping geometry - not machine learning

WE HAVE

- + scientifically defined tail rule
- + pairwise Spearman association
- + complete neighborhood retained
- + explicit provenance

WE DO NOT HAVE

- x no trained filters
- x no backpropagation
- x no learned objective
- x no image labels required

We have too much information

Now the notation has a physical meaning

$$S(x, y, dx, dy)$$

x, y

where the focal center lives

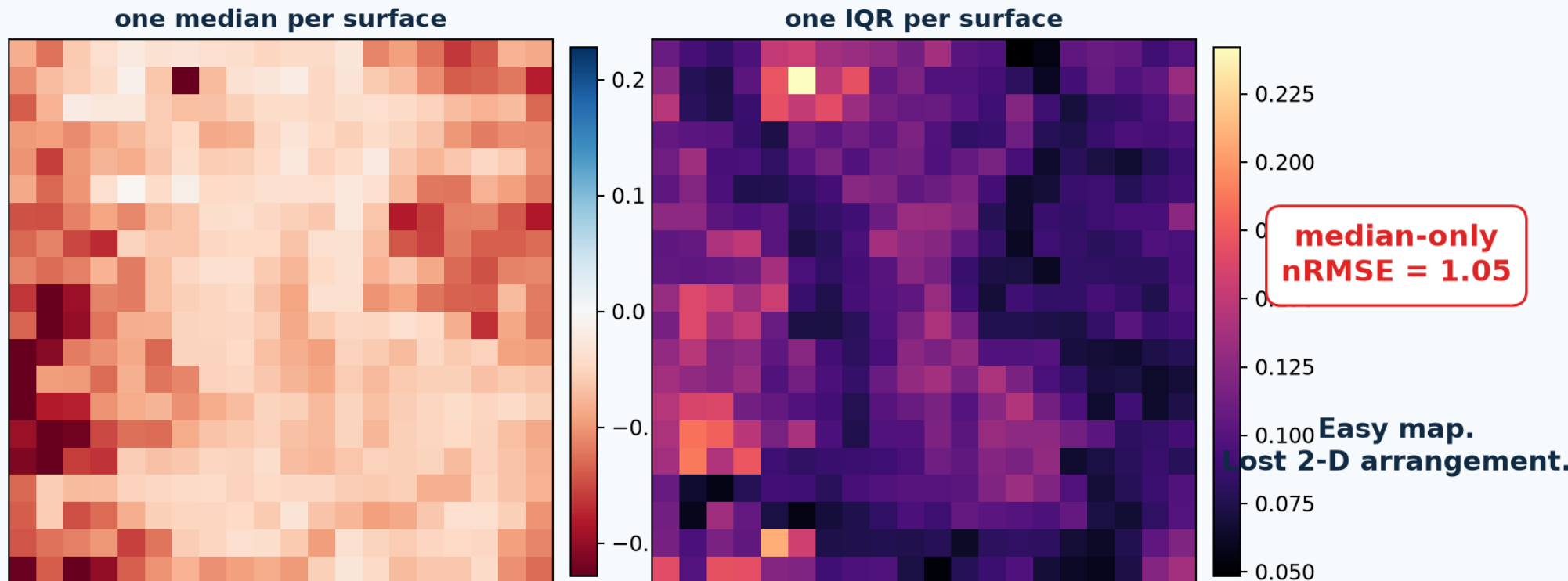
dx, dy

where the comparison lies
relative to that center

**A normal map has two spatial axes.
This field has focal geography plus within-surface geometry.**

The tempting easy solution

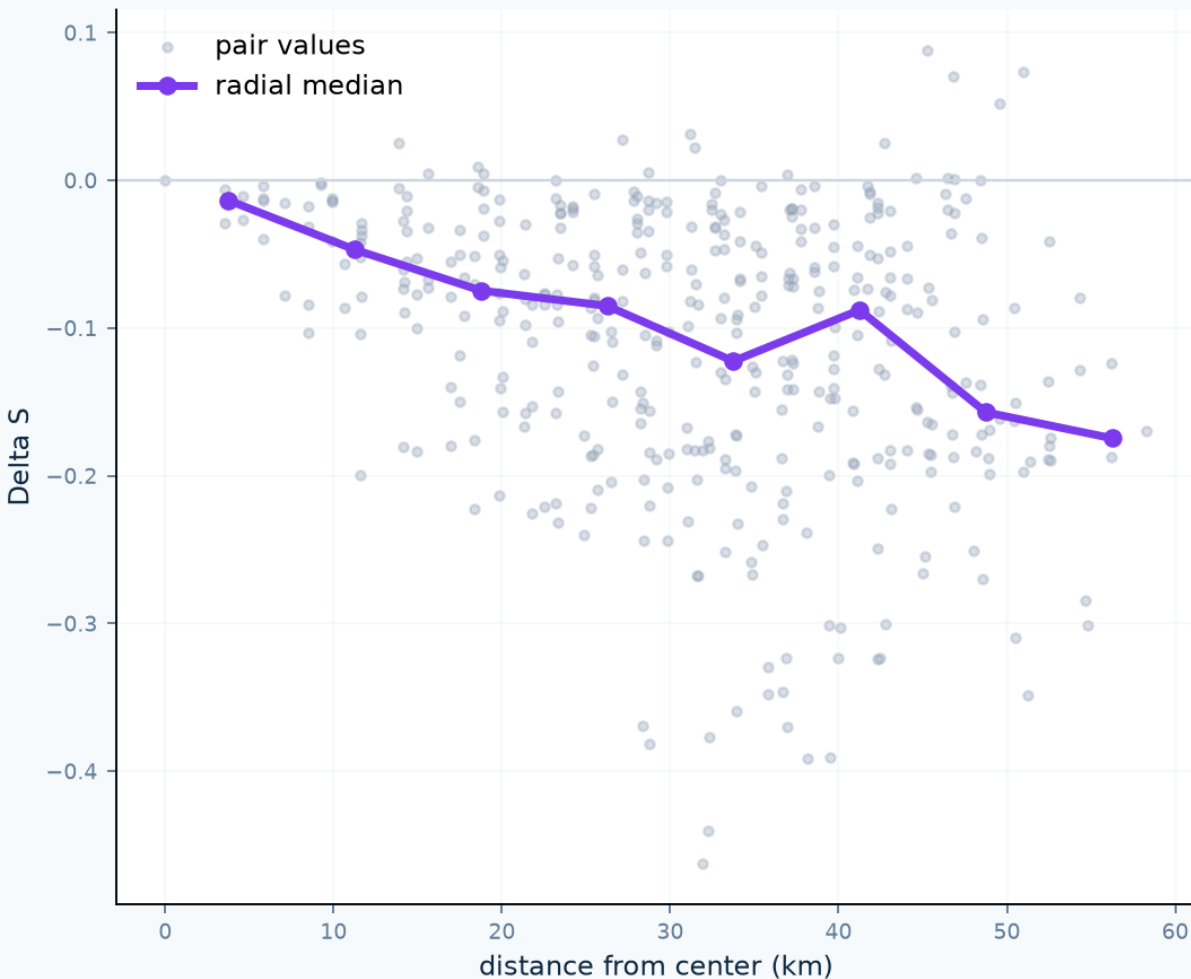
Collapse each surface independently to a median, mean, or spread statistic



Reduction is necessary. Independent scalar reduction is not automatically sufficient.

Radial reduction

Distance profiles retain structure, but they privilege radius



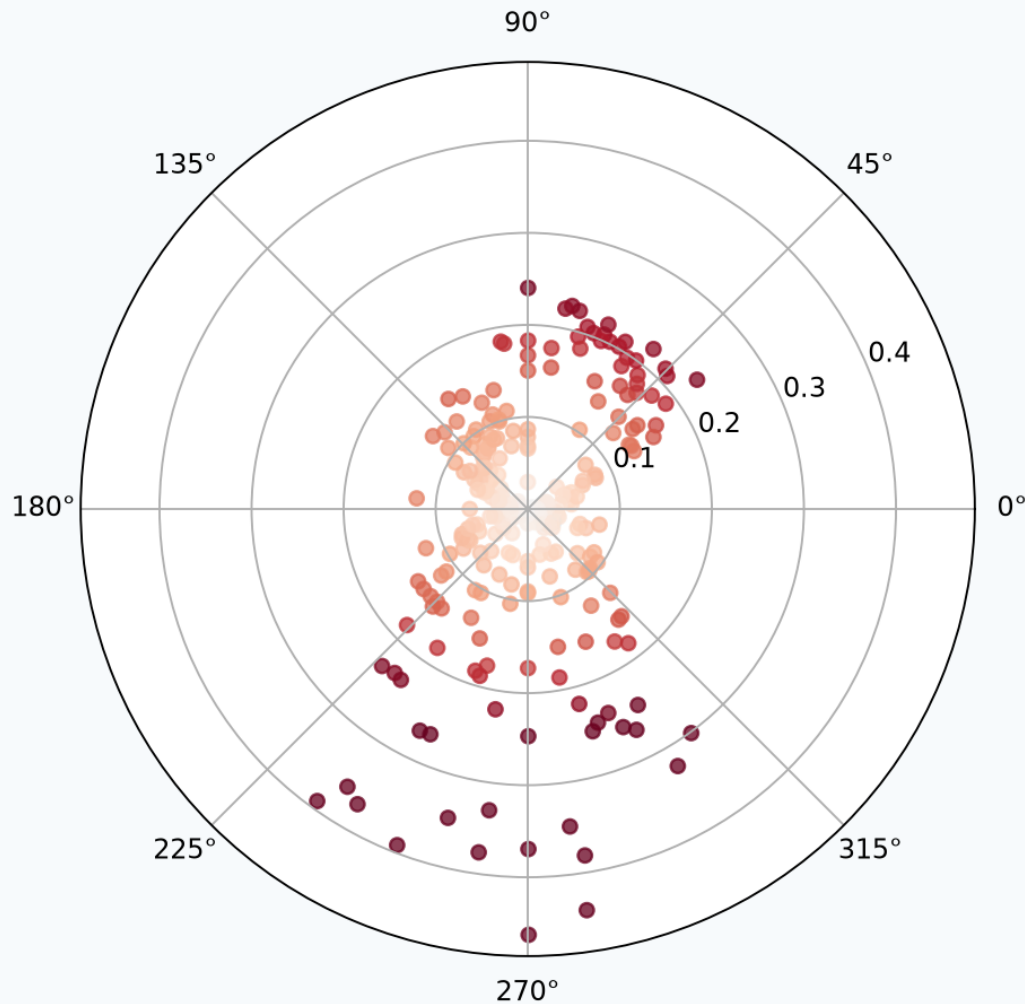
fine radial
nRMSE = 1.01

Useful baseline
not a law of
distance decay

Radial + directional

Direction preserves substantially more local geometry than median-only summaries

direction remains visible



median only



nRMSE 1.05

fine radial



nRMSE 1.01

radial + directional

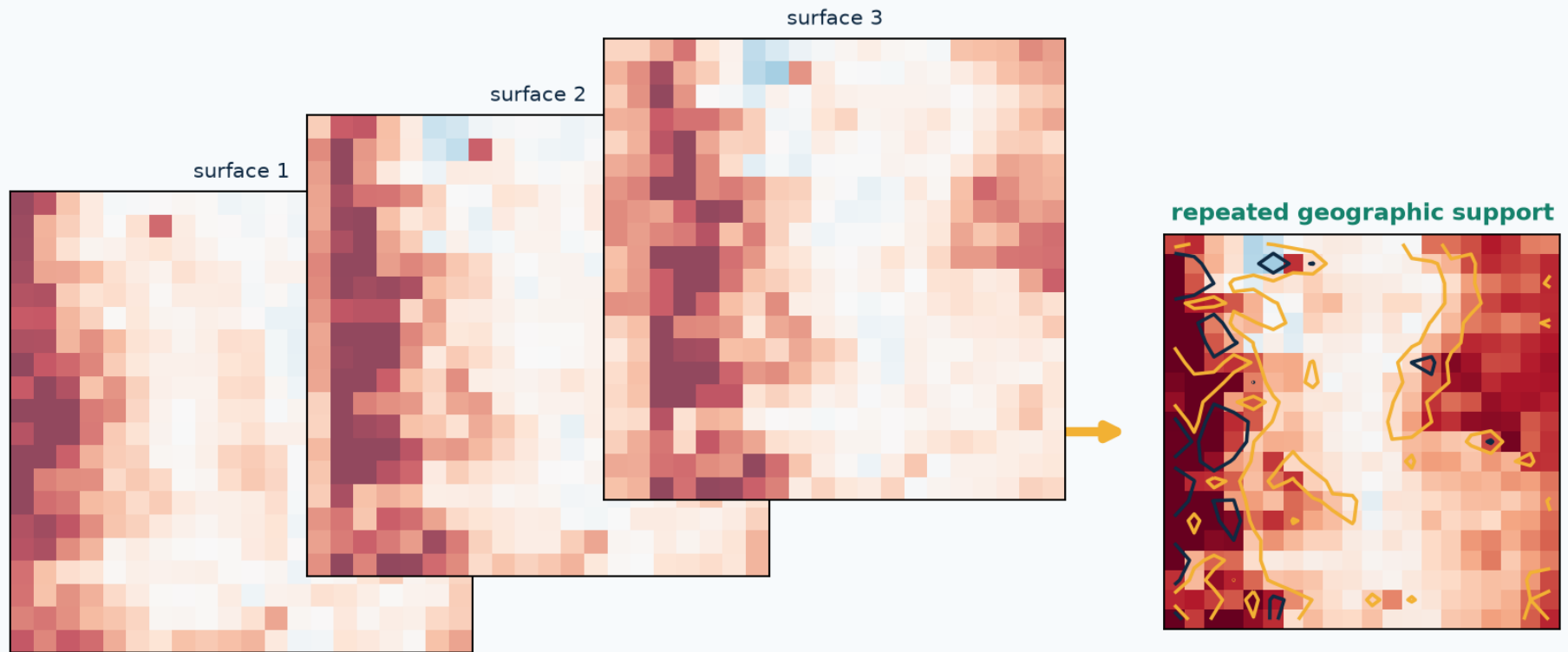


nRMSE 0.68

Lower is better. Each surface is still summarized independently.

The overlap idea

Instead of compressing each surface alone, align neighboring surfaces first

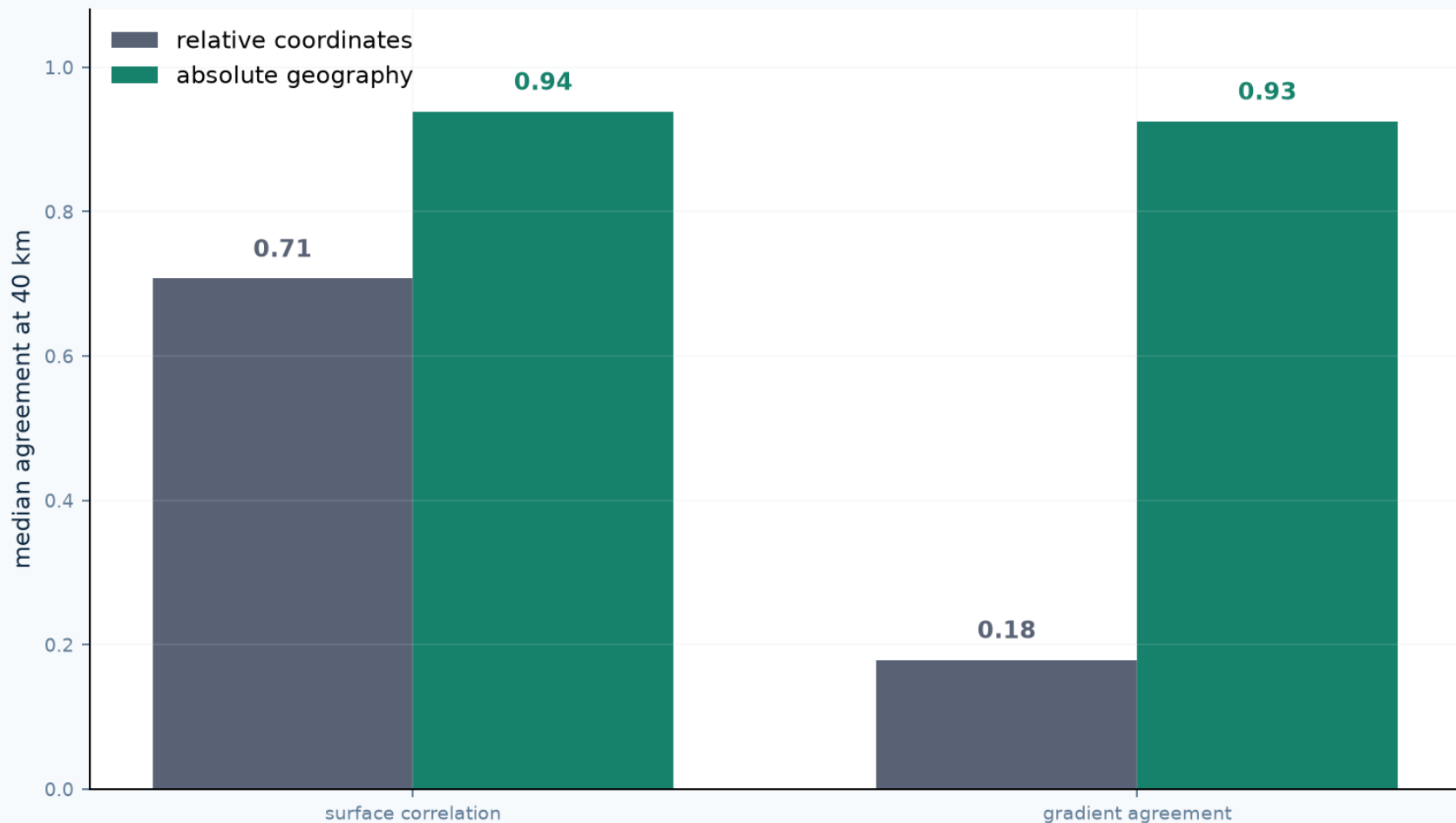


ALIGN TO GEOGRAPHY

THEN REDUCE

What the feasibility experiment found

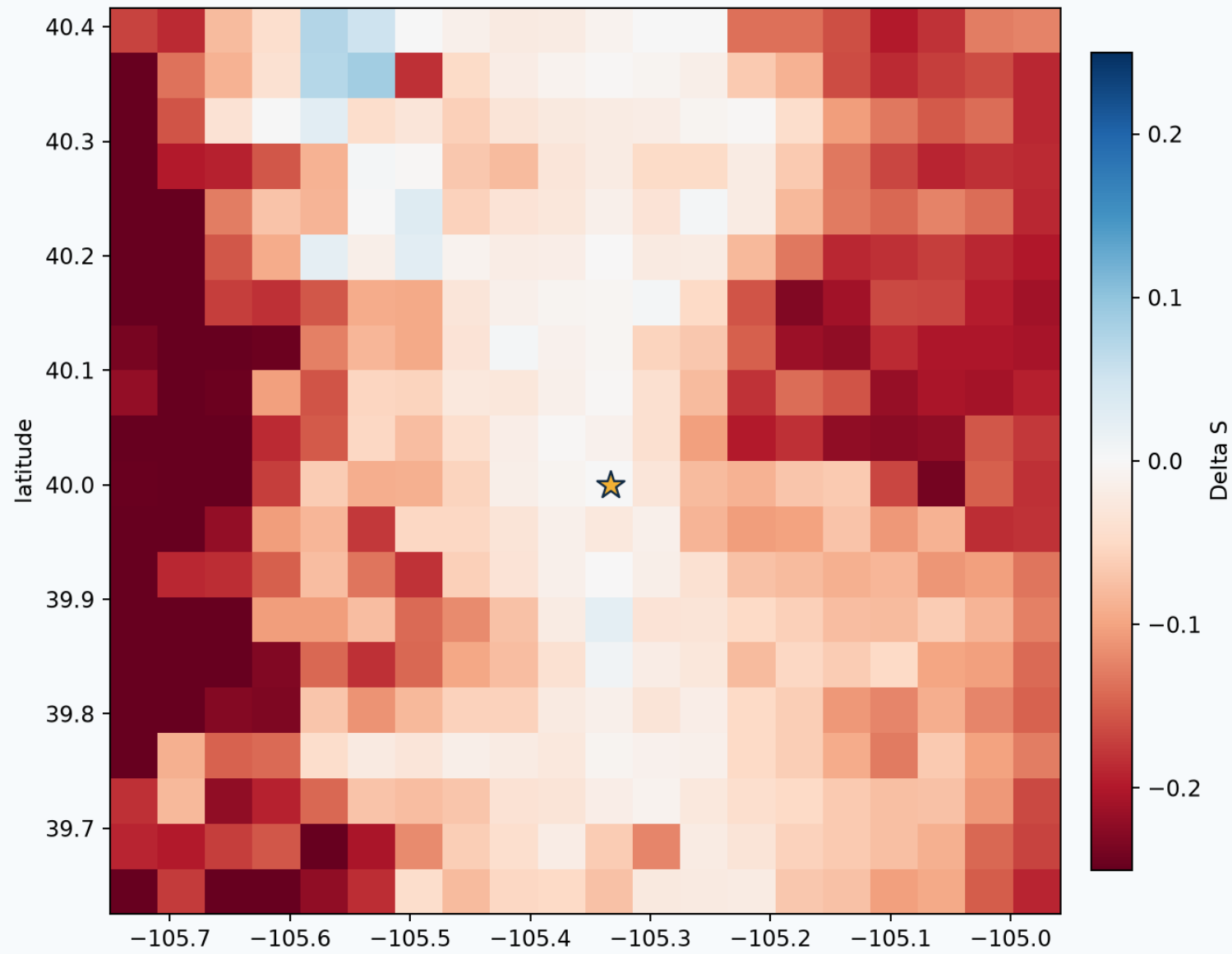
One bounded real block suggests recoverable geography at 40 km



Evidence for a direction - not a final overlap operator and not a production map.

Local block

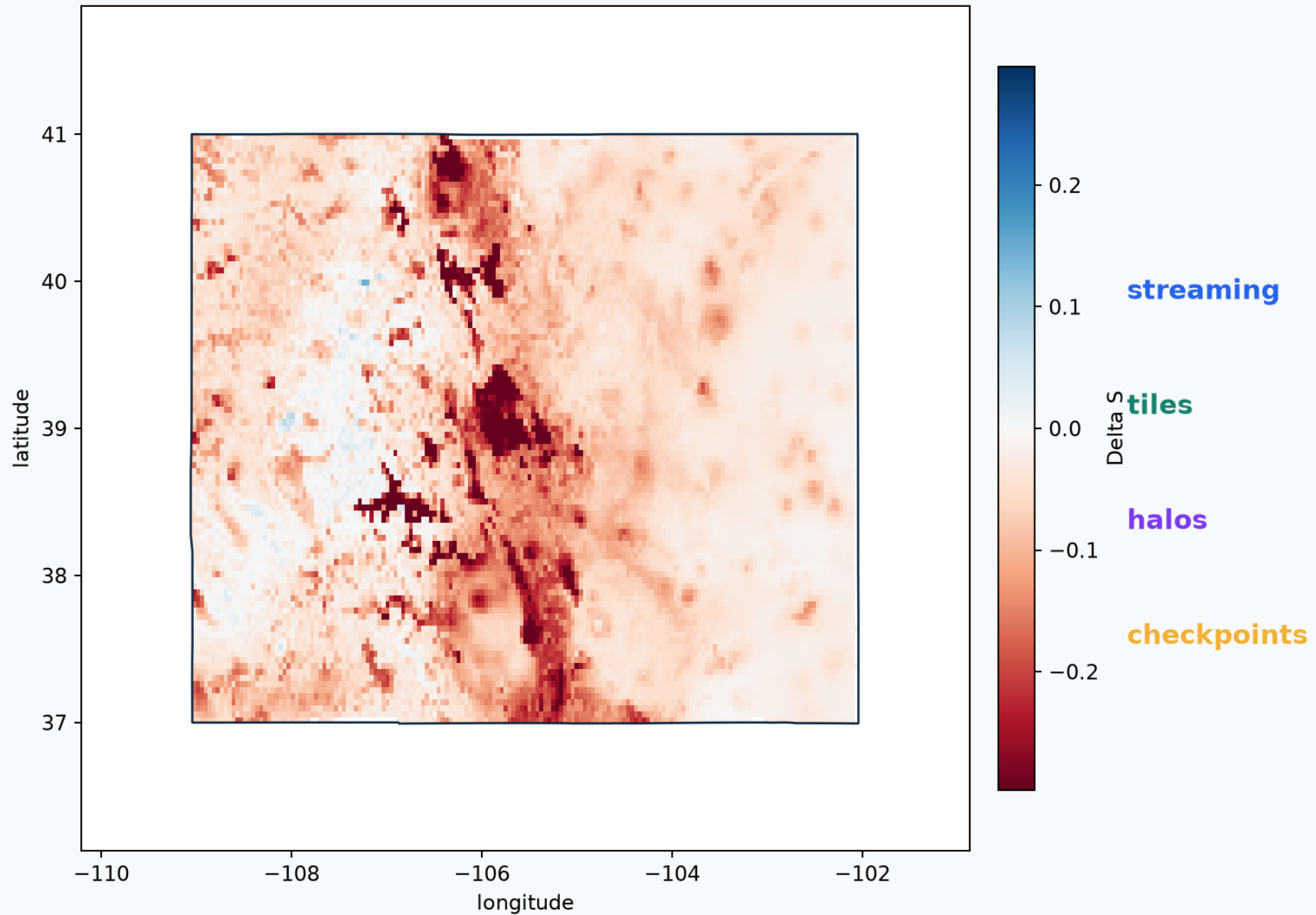
The current overlap experiment is a 20 x 20 Front Range relational field



Colorado

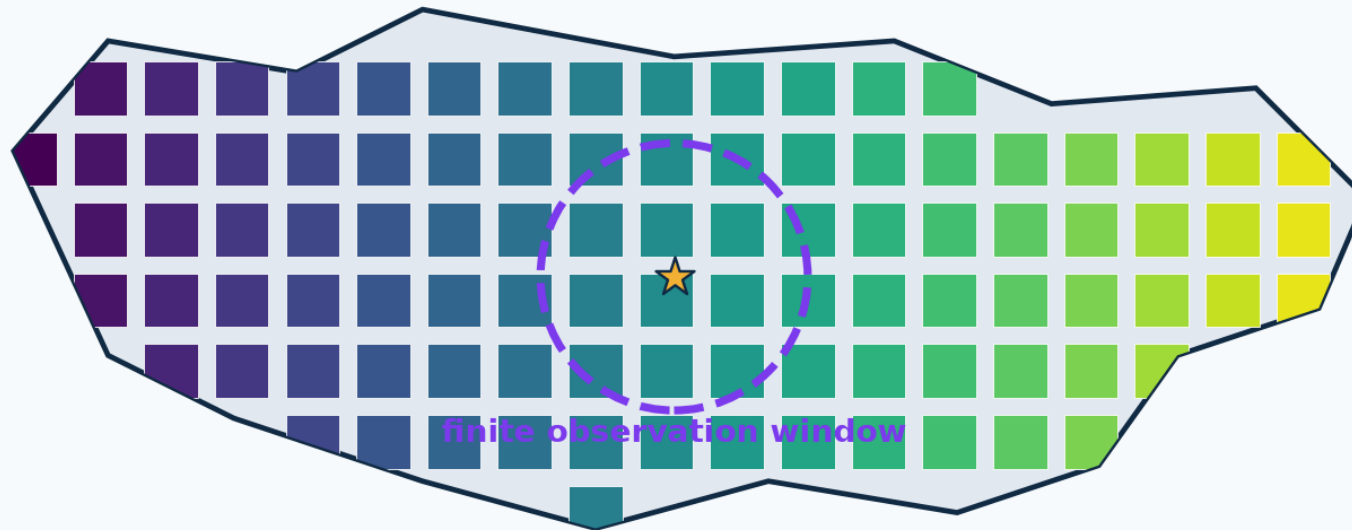
The same pair operation already runs statewide with tiles and halos

existing Delta median signature at 50 km



CONUS

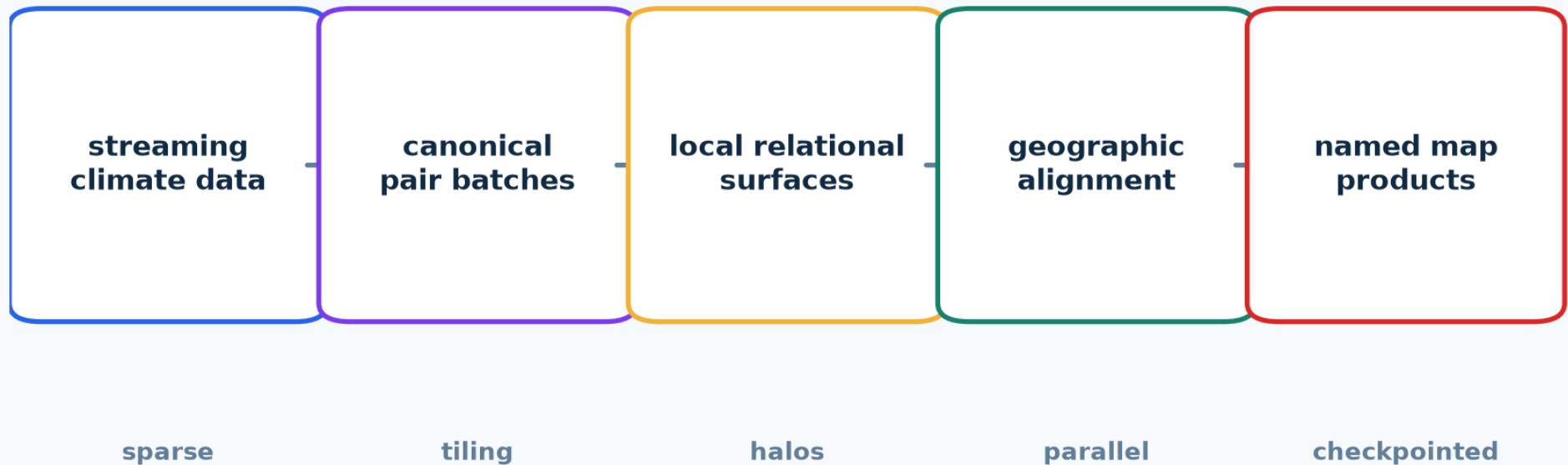
Every pixel can be a focal center with a finite surrounding observation window



Canonical pair identity avoids calculating $S(i,j)$ and $S(j,i)$ as separate evidence.

The computational picture

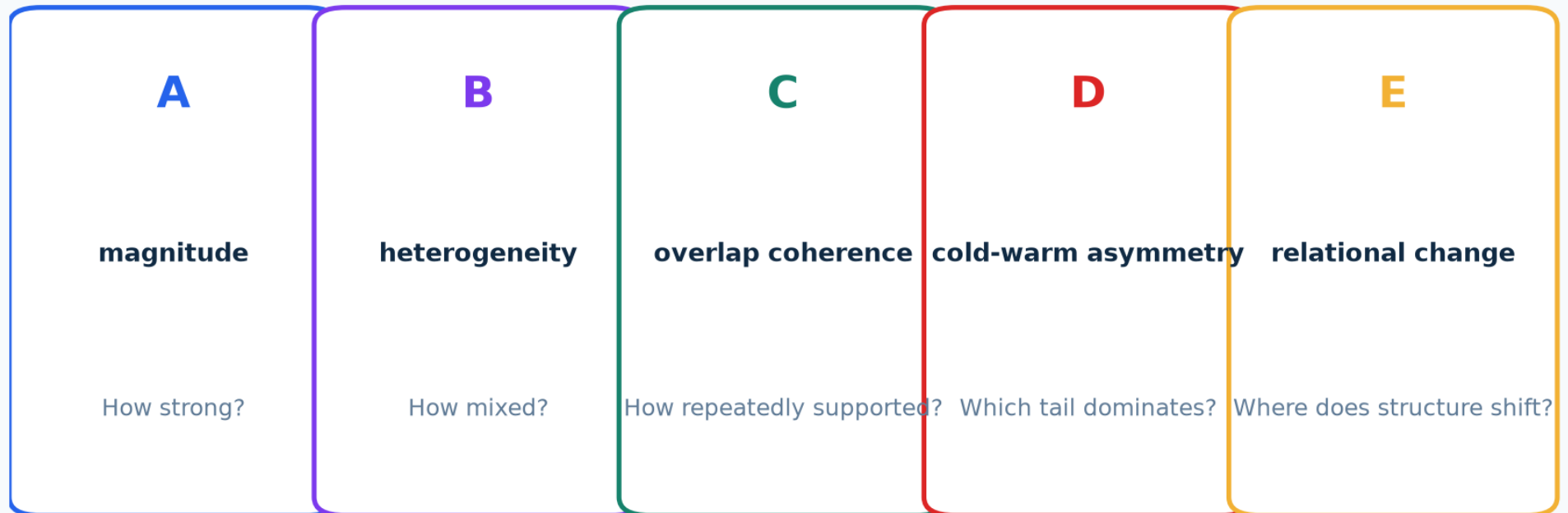
We do not need to hold the full relational field densely in memory



Dense $S(x,y,dx,dy)$ is a scientific concept - not a memory requirement.

What would the final map mean?

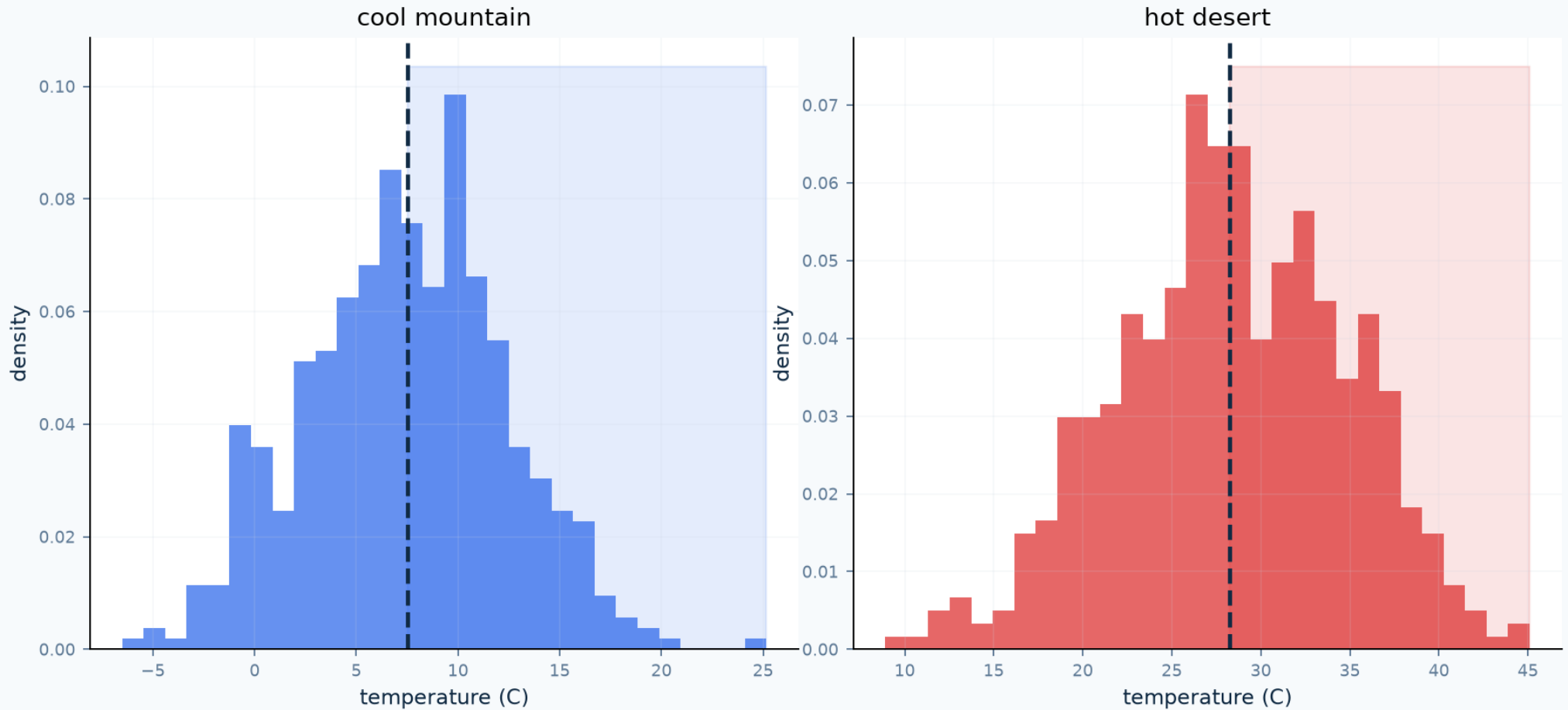
Candidate products answer different scientific questions



Do not collapse these into one unnamed 'synchrony score.'

What does 'hot' mean?

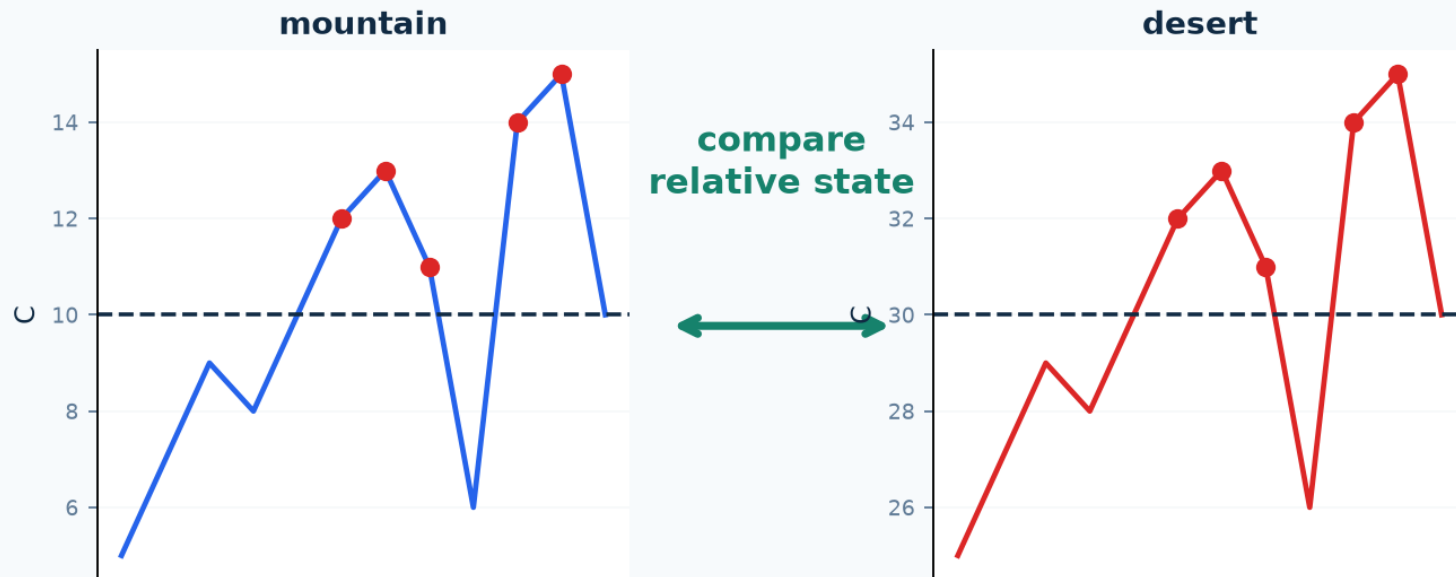
A temperature can be extreme in one climate and ordinary in another



Current implementation: each location uses its own pair-valid median inside the analysis window.

Relative extremes are a feature

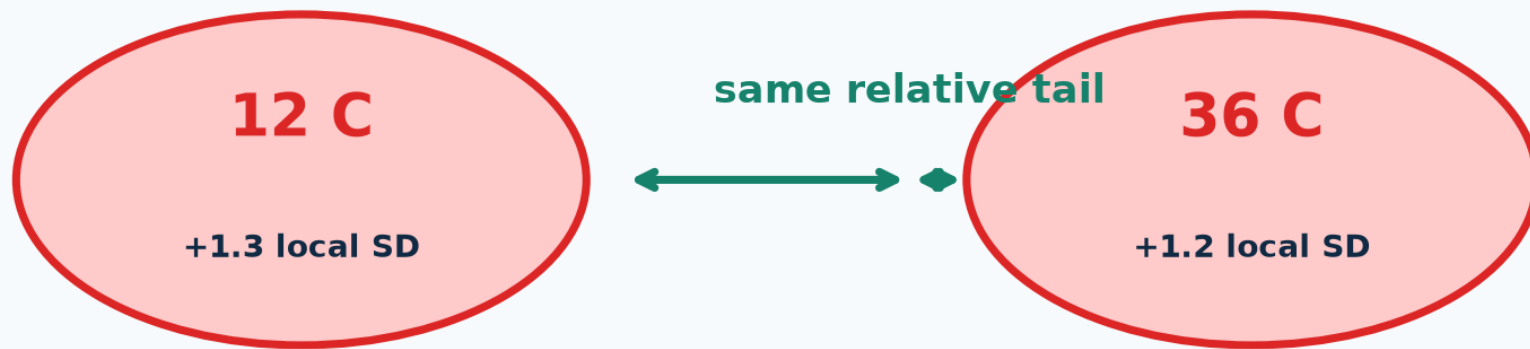
Different climates can both be unusually warm relative to themselves



Question enabled: Are both places unusually warm or cold relative to themselves at the same time?

But relativity creates a new problem

Relative synchrony does not necessarily mean absolute climatic similarity

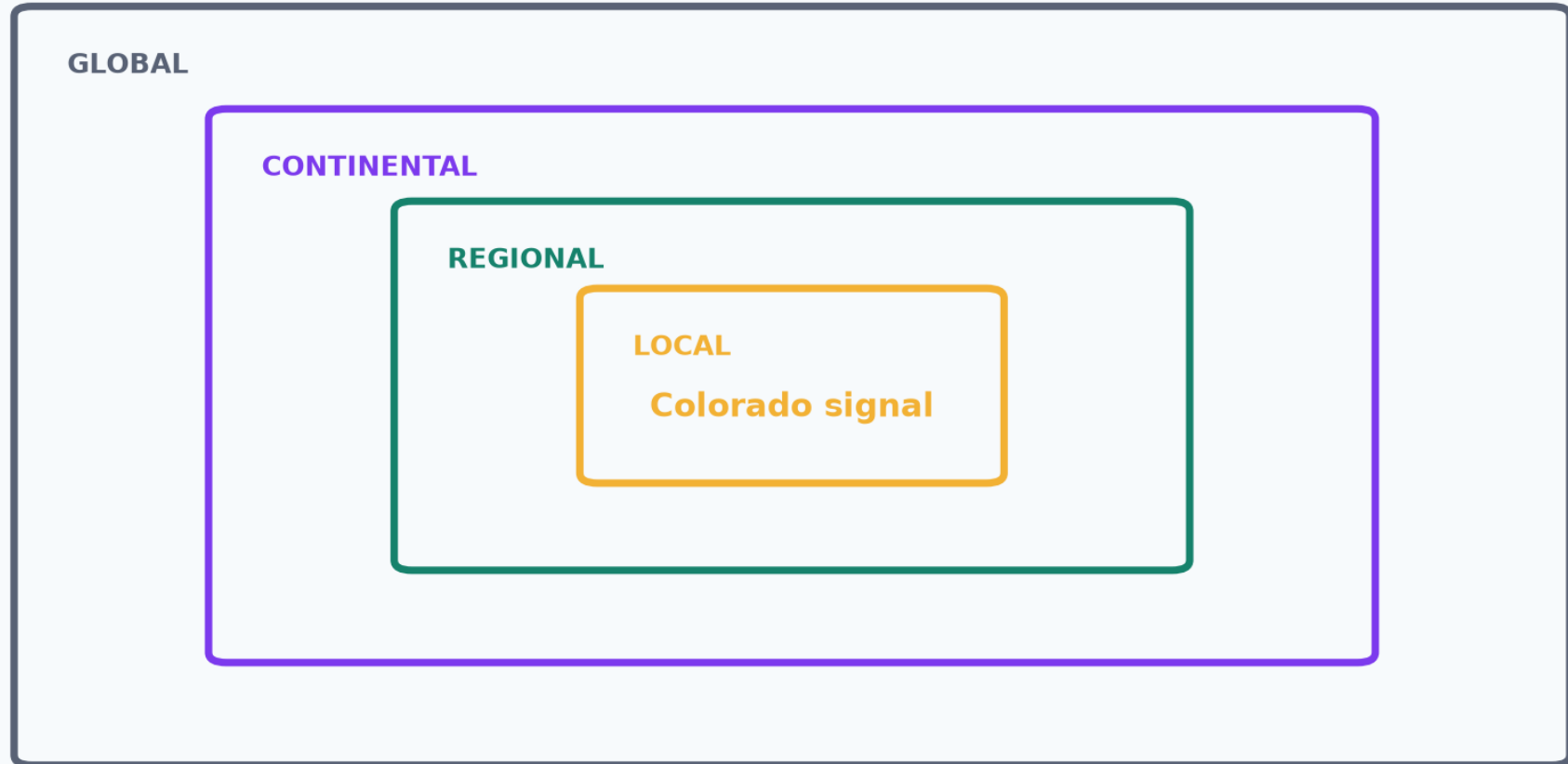


Very different absolute temperatures, variances, seasons, and climate regimes.

Keep the distinction: relative-extreme synchrony != absolute climate similarity

Local versus global variance

A global scale can wash out subtle but coherent regional structure



**Do not let Sahara-vs-Arctic variation set the only scale
for Front Range-vs-Western Slope structure.**

The normalization question

Five strategies for discussion - none is selected here

A**LOCAL RELATIVE**

comparable extremes; may hide absolute context

B**REGIONAL**

climate-aware; depends on region definitions

C**MULTISCALE**

preserves nested departures; more complex

D**LOCAL + ABSOLUTE**

keeps both meanings; two-axis product

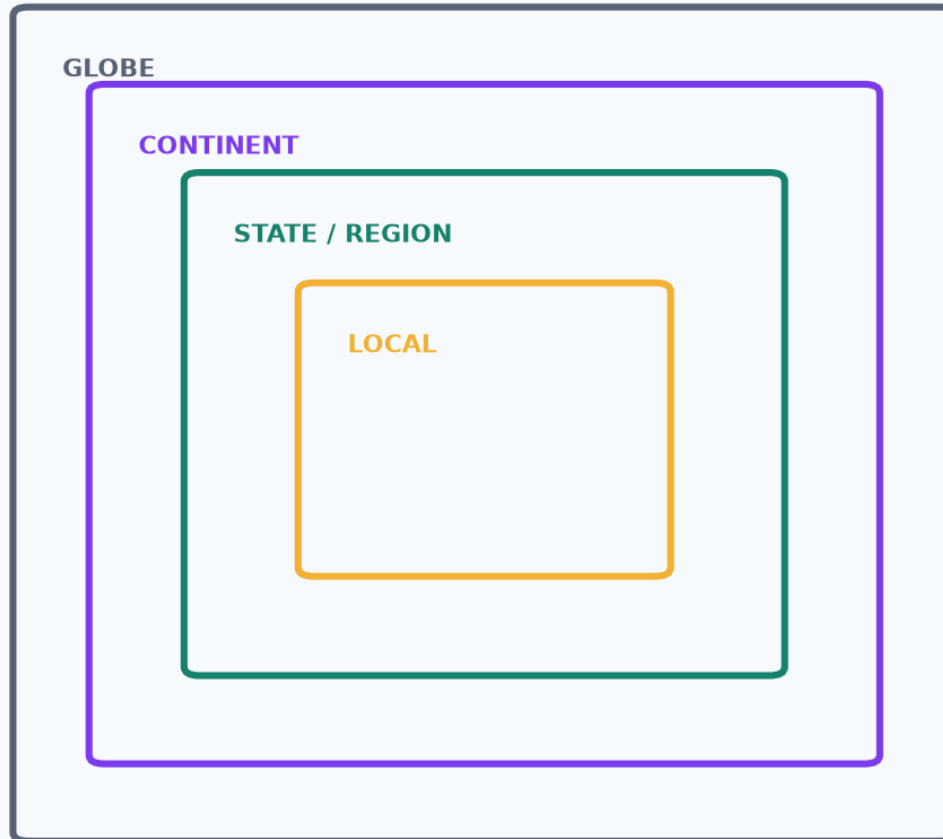
E**HIERARCHICAL**

separates levels; modeling assumptions

**CHOICE =
SCIENTIFIC QUESTION**

The answer can depend on the reference frame

Temporal reference, spatial reference, window, and normalization must be explicit



REFERENCE DISTRIBUTION

changes what counts
as unusual

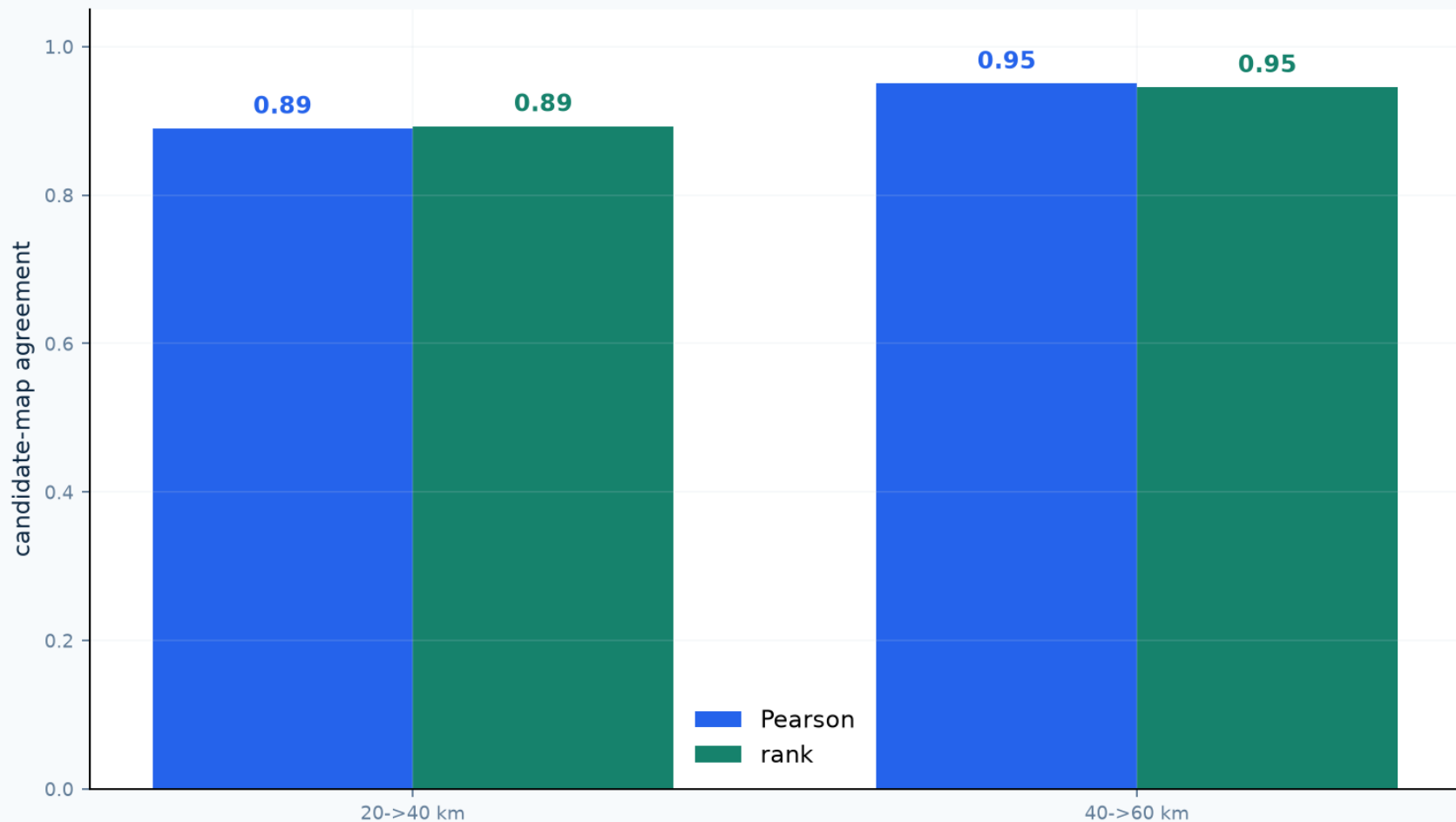
OBSERVATION WINDOW

changes which relationships
are visible

A synchrony product is not scale-free.

We should not search for one magic radius

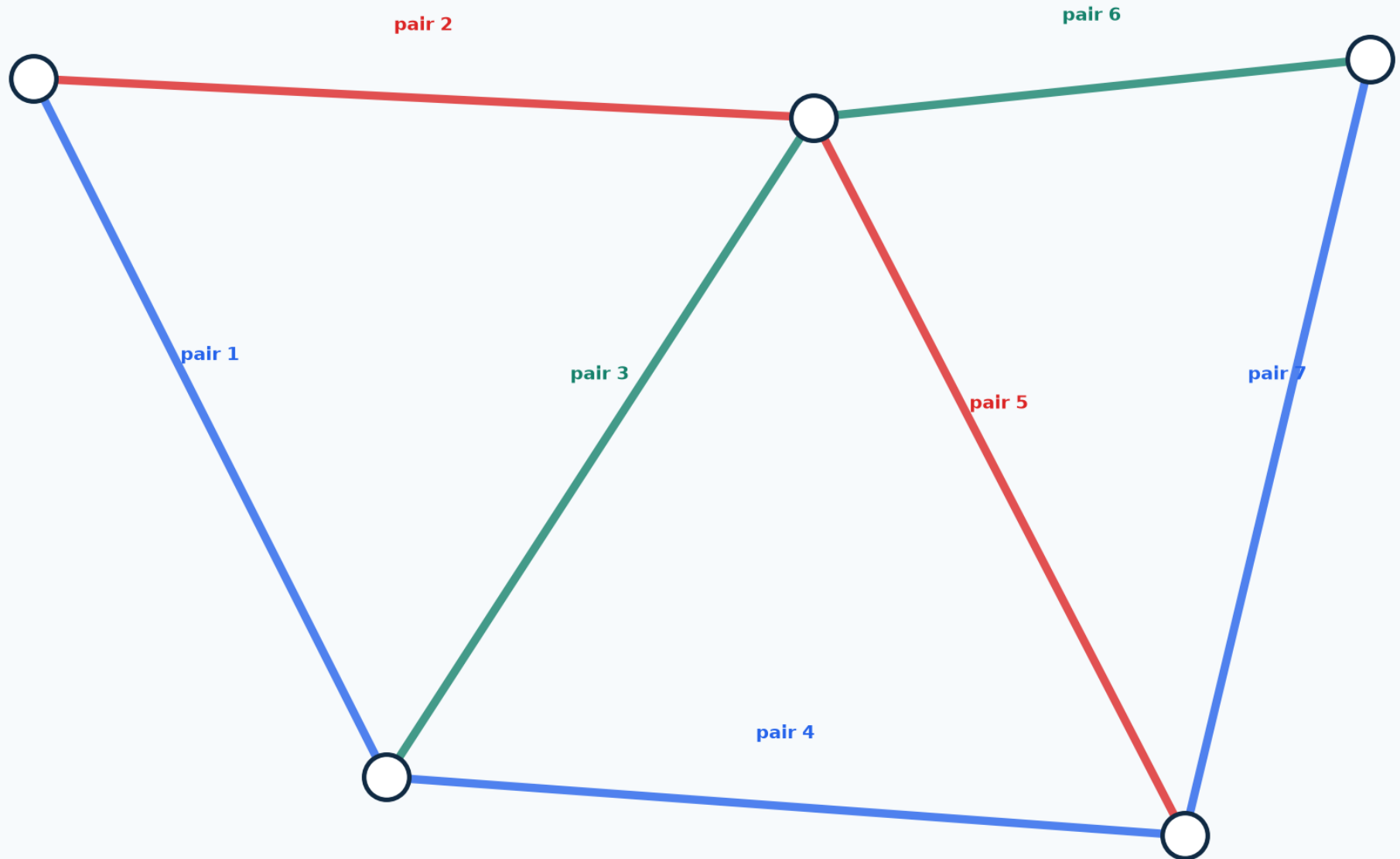
Rmax is an observation limit - not automatically the scale of synchrony



Ask whether geographic organization persists as the observation window expands.

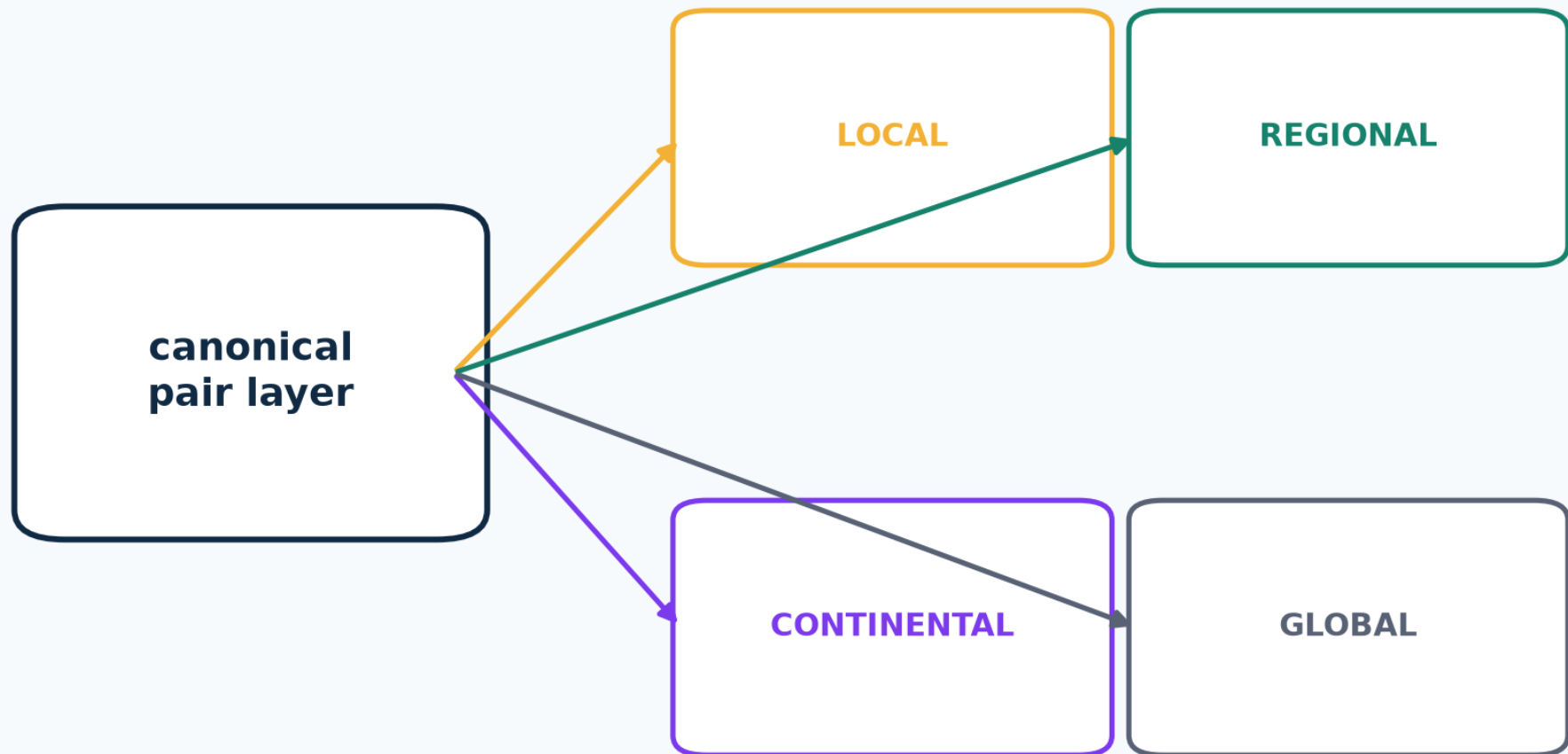
Keep relationships first

Canonical pair relationships are the durable scientific object



Then ask questions at multiple scales

Derive views from the same pair layer whenever definitions permit

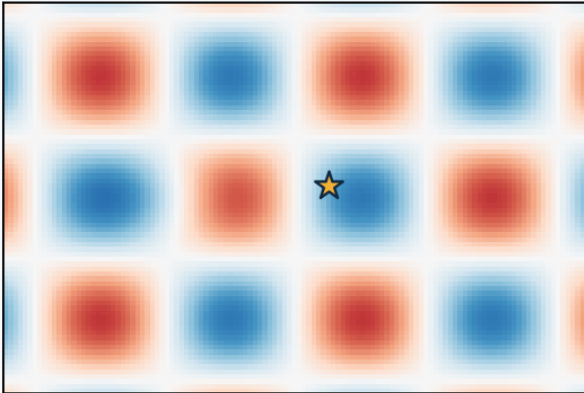


Reuse relationships; revisit threshold definitions when the scientific reference changes.

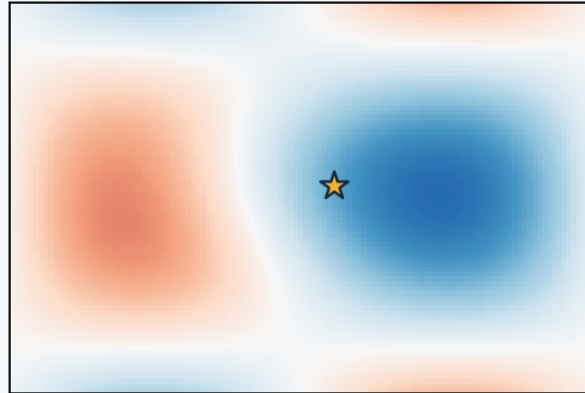
A multiscale synchrony map

One location may need local, regional, and broader-scale values

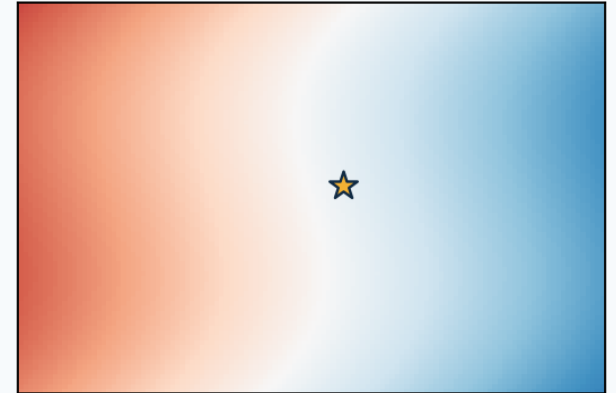
LOCAL VIEW



REGIONAL VIEW



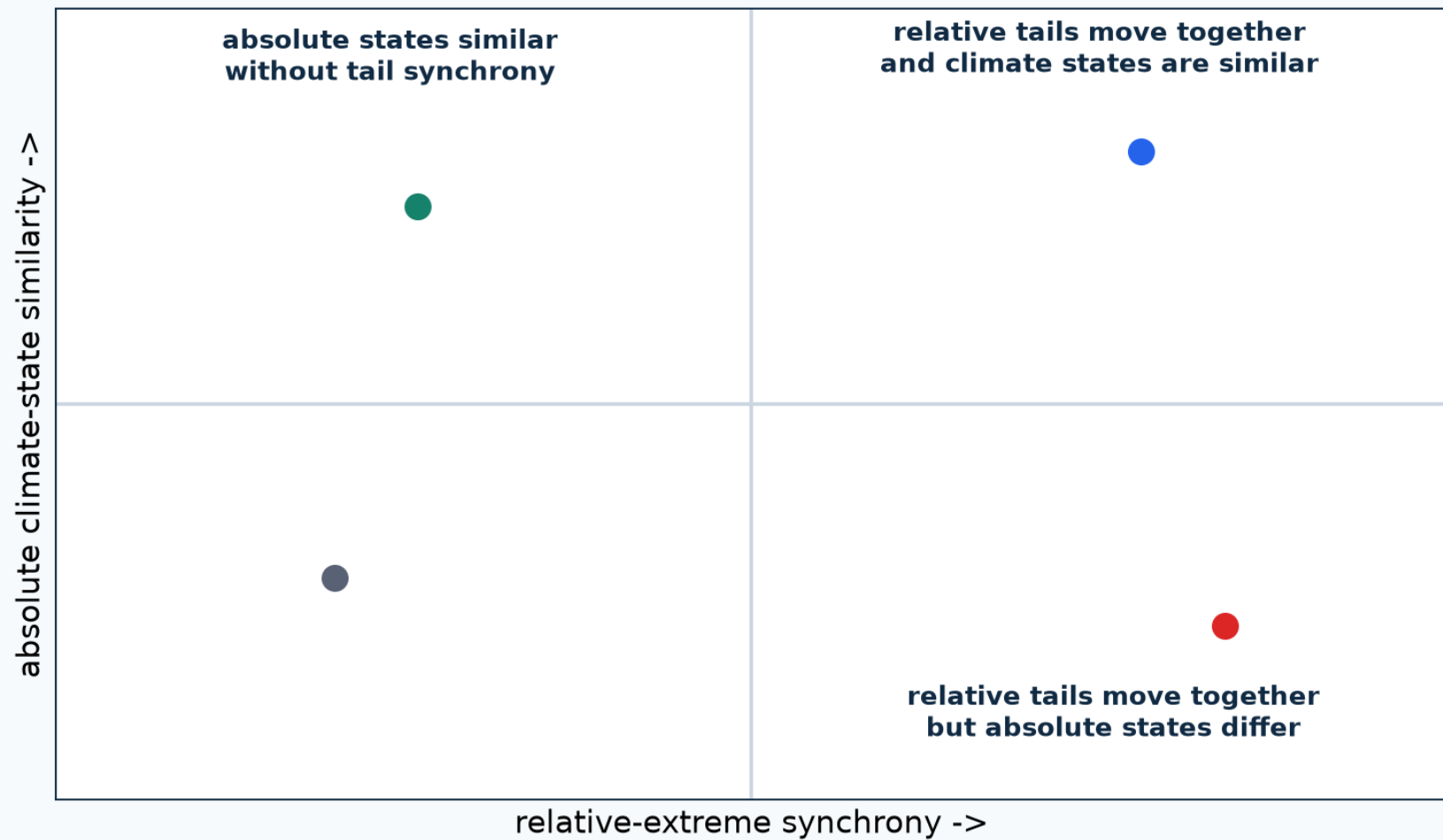
CONTINENTAL VIEW



Hypothesis for discussion - these scale-indexed maps are not yet implemented products.

Relative + absolute

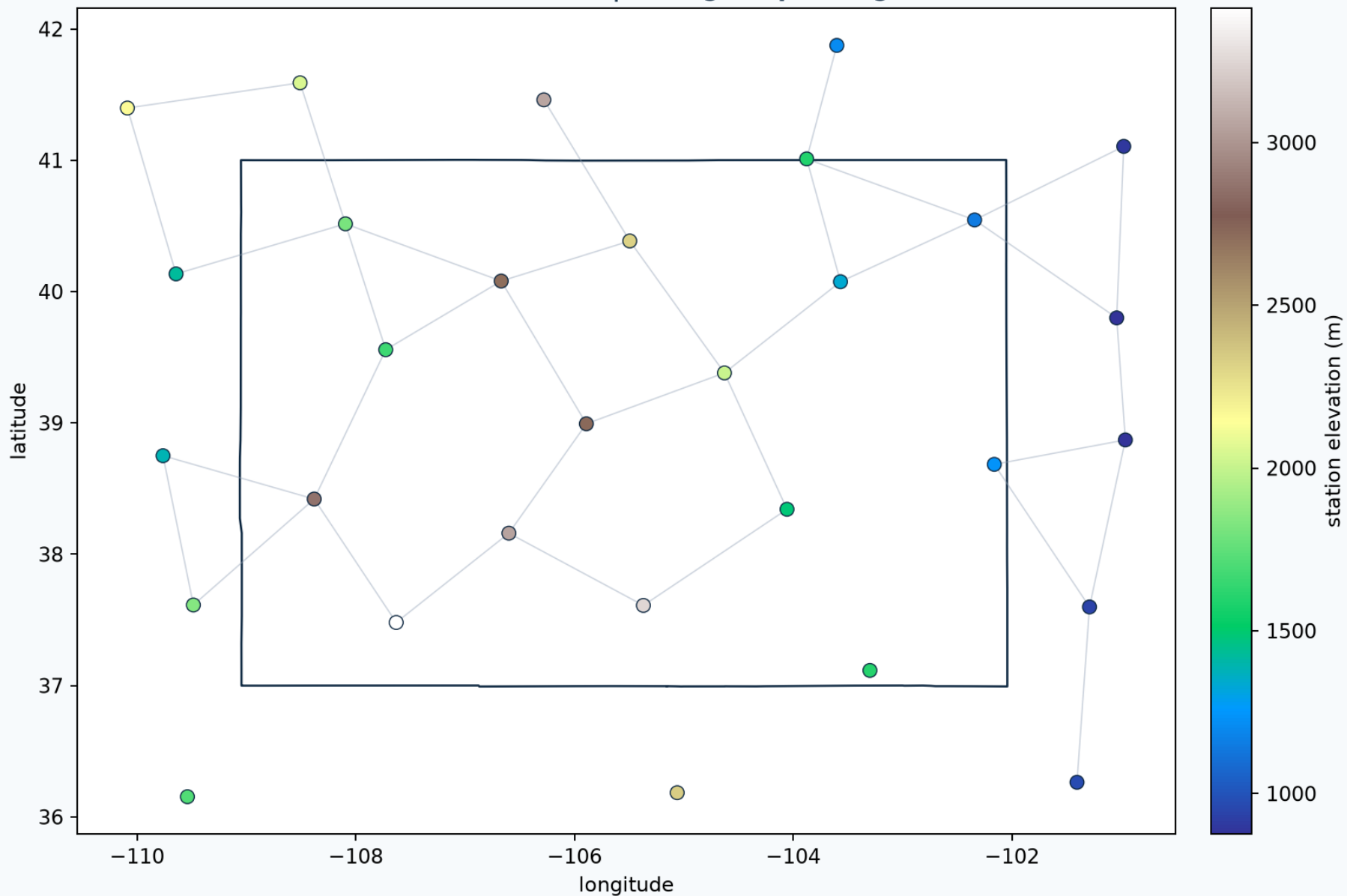
A two-axis view may prevent synchrony from being mistaken for climatic similarity



PRISM is a gridded model

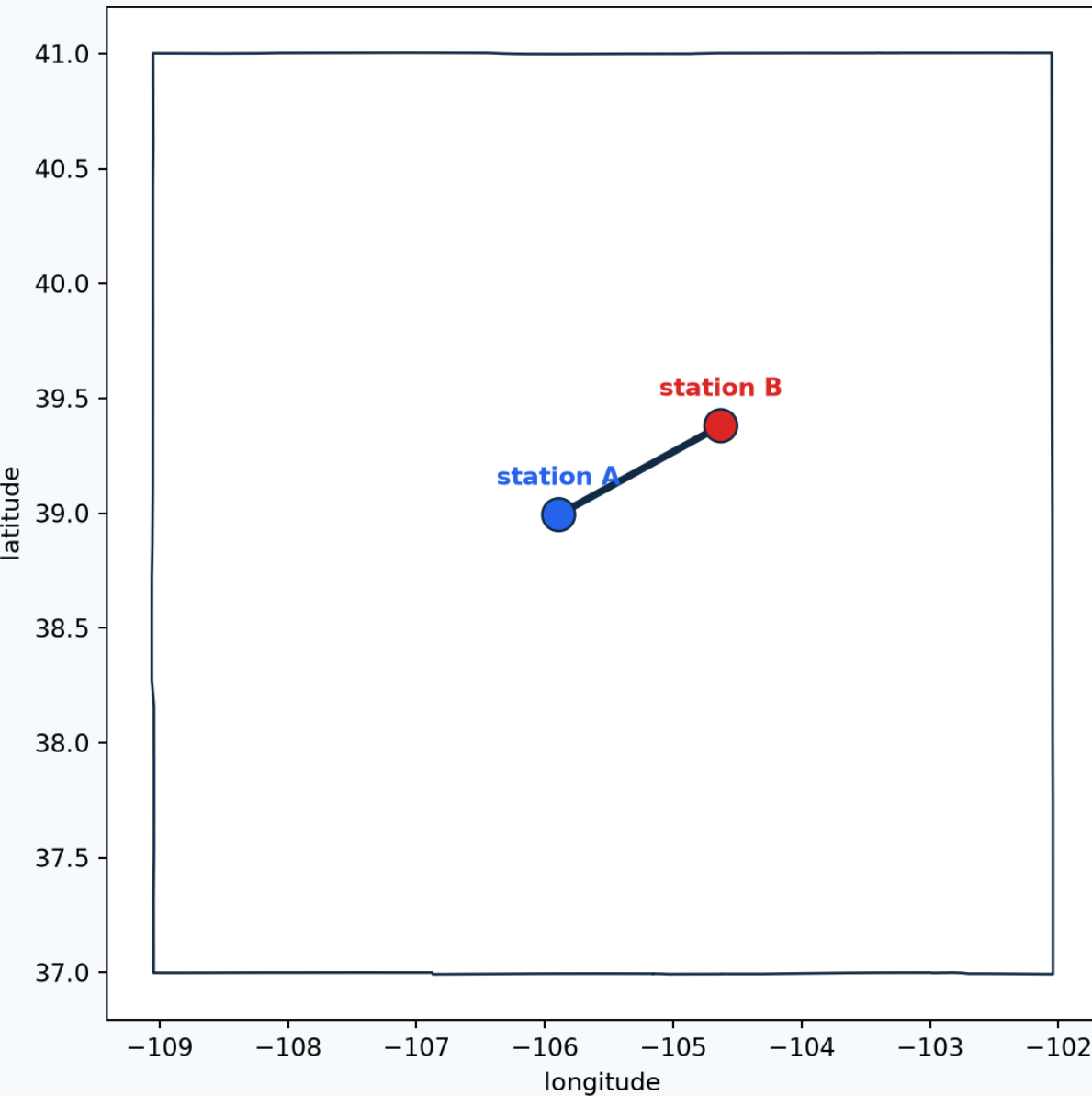
Raw GHCN-Daily stations provide another source of evidence

30 selected stations | 35 eligible pair edges



Build the same relationships from stations

Irregular station pairs use the SAME cold, warm, and Delta definitions



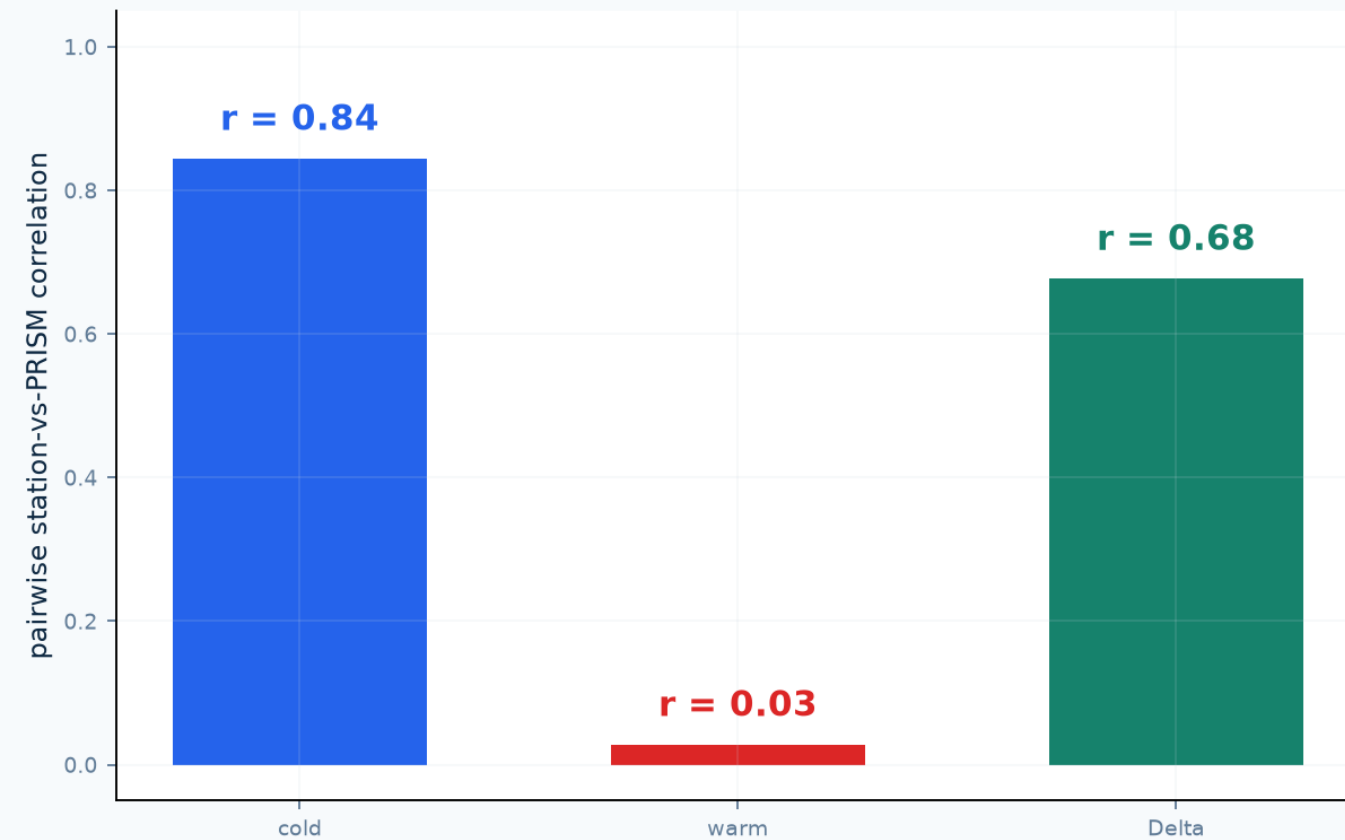
cold -0.16

warm 0.78

Delta -0.95

Current preliminary result

Descriptive station-vs-PRISM pair agreement - not independent validation



ENCOURAGING

cold + Delta

WEAK

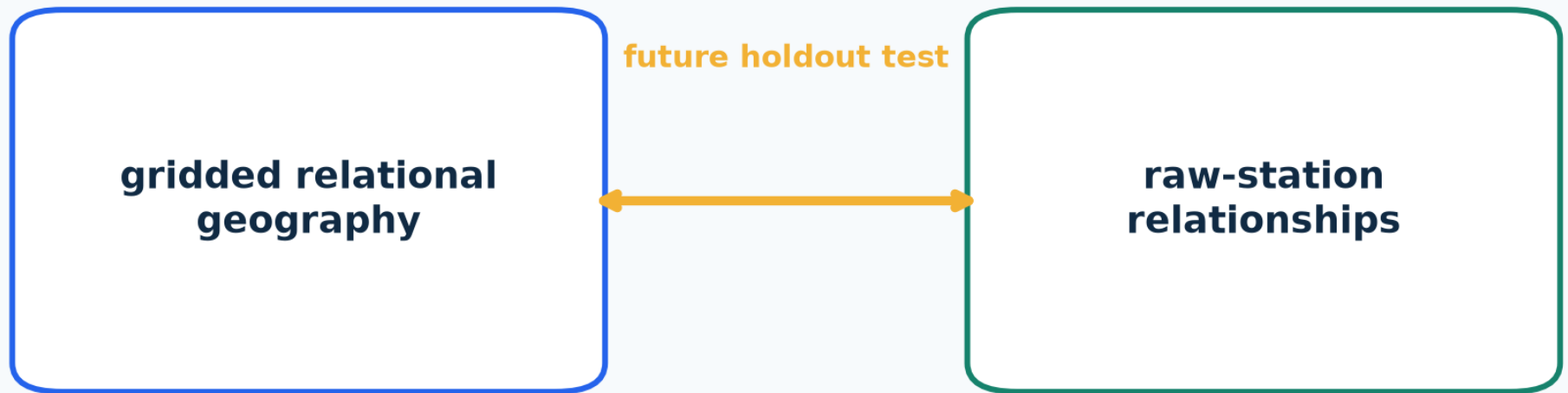
warm

LIMITS

sparse graph
shared stations
PRISM contribution
status: UNKNOWN

Why stations matter

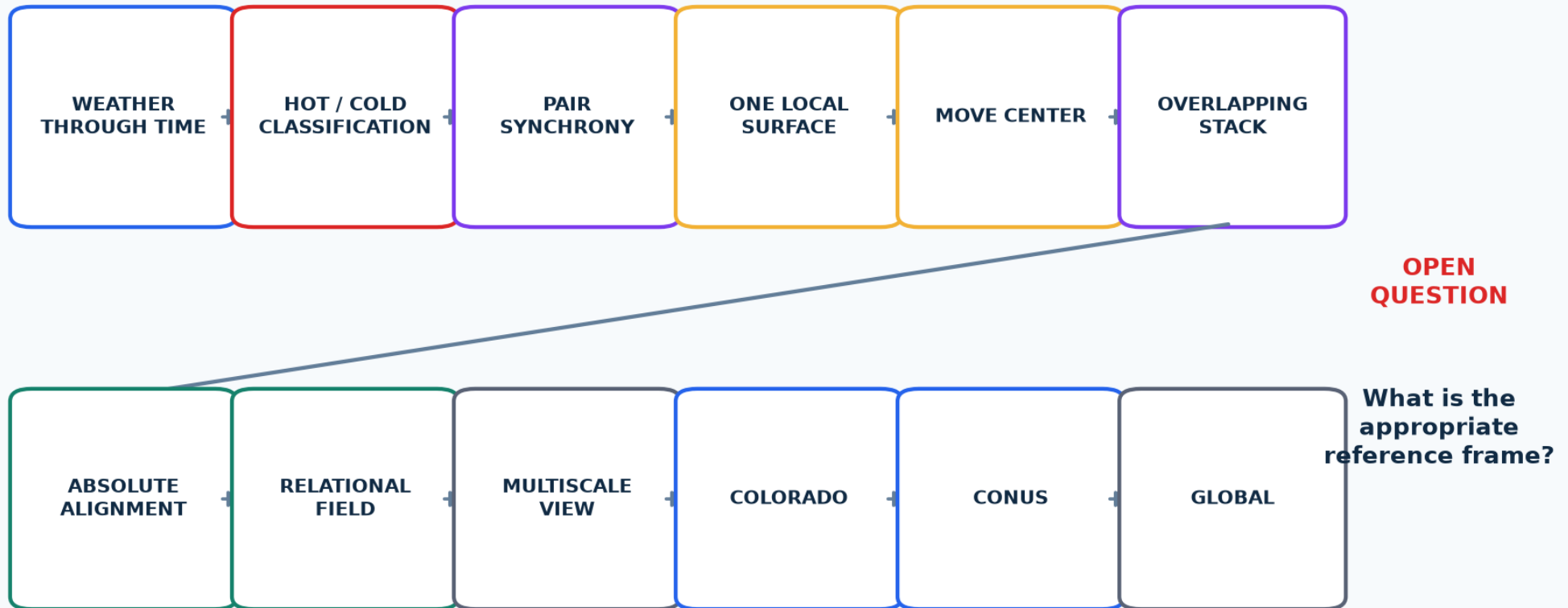
They offer an external observational test for at least some gridded structure



**Needed: temporal replication + leave-one-station-out + spatial-block holdout
+ product independence audit**

The entire method on one page

Every arrow preserves a scientific meaning - the reference frame remains open



What we know versus what we are deciding

Evidence and open design choices belong in separate columns

CURRENTLY SUPPORTED

- + pairwise tail calculation
- + local surface construction
- + surfaces overlap
- + absolute alignment helps
- + scalar reductions lose geometry
- + canonical symmetry

STILL DECIDING

- ? second-stage operator
- ? primary map quantity
- ? local / regional / global views
- ? global normalization
- ? window strategy
- ? validation sufficiency
- ? weak warm agreement

Question 1

Group discussion

WHAT SHOULD ONE PIXEL ON THE FINAL MAP MEAN?

average synchrony

local relational coherence

cold

warm

Delta

multiple named values

Question 2

Group discussion

WHAT SHOULD 'HOT' AND 'COLD' MEAN GLOBALLY?

pixel-relative

region-relative

globally standardized

multiscale

Question 3

Group discussion

HOW MUCH LOCAL VARIATION MUST WE PRESERVE?

If a global view is stable but hides meaningful regional structure, is that acceptable?

Which regional patterns must the method preserve by design?

Question 4

Group discussion

IS THERE ONE SYNCHRONY MAP OR A FAMILY OF MAPS?

cold x local / regional / global

warm x local / regional / global

Delta x local / regional / global

Question 5

Group discussion

WHAT WOULD CONVINCE US THE MAP IS REAL?

observation-window stability

temporal replication

raw-station support

synthetic recovery

independent climate products

interpretable geography

We now understand the object well enough to decide how it should scale.